

Why Rotating Fluids Make Polygons

Paper II: Classical and Quantum Physics

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Abstract

We prove that the regular N -gon of point vortices is a constrained energy minimum on the angular-impulse surface for $N \leq 7$, resolving the tension between Thomson stability and Onsager’s statistical-mechanics picture. The $N = 7$ marginal case is resolved by an exact quartic normal form ($\alpha_0 = 45/14$), and the result extends to continuous vorticity via a profile-independent blob bridge at $O(\varepsilon^2)$. Saturn’s oblateness ($f = 0.098$) reverses the hexagonal eigenvalue from -0.12 to $+0.01$, determining whether the hexagon exists. A constraint-intersection principle unifies the Rossby, Thomson, and packing bounds: energy monotonicity $H_{N+1} > H_N$ selects the maximum stable polygon, explaining Saturn ($N = 6$, Rossby-limited), Jupiter north ($N = 8$, Thomson-limited), and Jupiter south ($N = 5$, packing-limited). Predictions for Neptune ($N \in \{3, 4, 5\}$) and Uranus ($N = 0$) are stated.

1 Introduction

The regular N -gon of point vortices is simultaneously an unconstrained energy saddle and a constrained energy minimum (Corollary 3.3). This paper proves the latter for $N \leq 7$ (Theorem 3.2), resolving the tension between Thomson–Havelock linear stability and Onsager’s negative-temperature prediction of large-scale clustering. The marginal $N = 7$ case is resolved by an exact quartic normal form with $\alpha_0 = 45/14$ (§5). Paper I provides the Havelock eigenvalue identity used throughout.

Five extensions establish that the result is not an artifact of the point-vortex idealisation. Central-vortex stabilisation (§4) explains Jupiter’s $N = 8$ octagon. The blob bridge (§7.1) shows that finite-core vortices inherit point-vortex stability. The Rossby bridge (§7.2) connects continuous jet meanders to the discrete energy landscape. The multi-ring dichotomy (§6) and deformation-radius corrections (§8) complete the robustness analysis. Planetary oblateness (§9) determines whether Saturn’s hexagon can exist, and the constraint-intersection principle (§10) unifies the three selection mechanisms into a single framework covering all four giant planets. The gravitational analogue is treated briefly in §13; the full gravitational theory is in Paper III.

2 Point vortex dynamics on a two-dimensional surface

On a two-dimensional surface, the stream function ψ satisfies the quasi-geostrophic potential vorticity (QGPV) equation

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + \beta y. \quad (2.1)$$

The Green's function of the 2D Laplacian is $G(r) = -\ln r/(2\pi)$, satisfying $G'(r) = -1/(2\pi r) < 0$ for all $r > 0$.

For N point vortices with equal circulations κ on the collision-free configuration space $\{|z_j - z_k| > 0 \text{ for all } j \neq k\}$, where H is smooth and $\mathbb{Z}_N$ -invariant, the interaction energy is

$$H = - \sum_{j < k} \ln |z_j - z_k|, \quad (2.2)$$

where we absorb $\kappa^2/(4\pi)$ into the normalization. By the equivariant critical point theorem, the regular N -gon—a fixed point of $\mathbb{Z}_N$ —is a critical point of H .

Point vortex dynamics conserves three quantities: the energy H , the angular impulse $L = \sum |z_k|^2$, and the linear impulse $\mathbf{P} = \sum z_k$. The physically accessible configuration space is the $(2N - 3)$ -dimensional surface $\{L = NR^2, \mathbf{P} = 0\}$.

2.1 The centered spiral deformation

Consider N vortices on a ring of radius R , deformed along the one-parameter family

$$z_k(\sigma) = R e^{\sigma(k-(N-1)/2)/N} e^{2\pi i k/N}, \quad k = 0, \dots, N-1. \quad (2.3)$$

At $\sigma = 0$: the regular N -gon. The index shift $k - (N-1)/2$ centers the spiral so that the reversal $k \mapsto N-1-k$ acts as $\sigma \mapsto -\sigma$, giving $H(\sigma) = H(-\sigma)$ and hence $H'(0) = 0$.

Remark 2.1 (Möbius origin). *The positions $\{z_k(\sigma)\}$ are the orbit of $z_0 = R$ under $z \mapsto \lambda z$ with $\lambda = e^{\sigma/N + 2\pi i/N}$. At $\sigma = 0$ the generator is elliptic (rotation by $2\pi/N$); for $\sigma \neq 0$ it is loxodromic. The spiral direction is the unique $\mathbb{Z}_N$ -equivariant perturbation tangent to a Möbius orbit. The dilation subgroup gives the N -body scaling identity $H(bz_1, \dots, bz_N) = H - \frac{N(N-1)}{2} \ln b$ (Paper I, Corollary 2.4).*

3 The constrained energy landscape

3.1 Energy curvature along the spiral

Theorem 3.1 (Energy curvature at the polygon). *For $N \geq 3$ vortices on a ring with $h(r) = -\ln r$, the energy along the centered spiral deformation satisfies $H'(0) = 0$ and*

$$H''(0) = -\frac{1}{4N^2} \sum_{m=1}^{N-1} \frac{(N-m)m^2}{\sin^2(\pi m/N)} < 0. \quad (3.1)$$

Proof. Part (a): $H'(0) = 0$ by the centering symmetry $k \mapsto N-1-k$. Part (b): Define $F(u, m) = \ln[1 - 2\cos(2\pi m/N) e^{um} + e^{2um}]$ with $u = \sigma/N$. Direct differentiation gives

$F''(0, m) = m^2/[2 \sin^2(\pi m/N)]$. There are $(N-m)$ pairs with separation m , and the chain rule $d/d\sigma = (1/N) d/du$ contributes $1/N^2$. This derivation uses the specific structure of $h = -\ln r$: the general formula Paper I, eq. 3.1 has two terms, but for the logarithmic interaction $(rh')' = (-1)' = 0$, so the S_m terms (first term in Paper I, eq. 3.1) vanish identically. Only the second term survives, giving (3.1) as a special case of Paper I, eq. 3.1 with $A_m = 0$ and $B_m = h'(d_m) m^2(N-m)/(N^2 d_m) = -m^2(N-m)/(N^2 d_m^2)$. Substituting $d_m = 2 \sin(\pi m/N)$ yields (3.1). Every factor in the sum is strictly positive, so $H''(0) < 0$. $\square$

3.2 The constrained energy minimum

By Theorem 3.2 (Paper I), the N -gon is a constrained energy minimum on the angular-impulse surface $\{L = NR^2, \mathbf{P} = 0\}$ if and only if $N \leq 7$. The Lagrangian Hessian is

$$\nabla^2 \mathcal{L} = \nabla^2 H - 2\mu_L I, \quad \mu_L = -\frac{N-1}{4}. \quad (3.2)$$

The following decomposition, used in the quartic computation (§5) and the blob bridge (§7.1), gives the mode-by-mode structure:

Proposition 3.2 (Hessian block structure). *The Hessian $\nabla^2 H$ at the N -gon block-diagonalises under the $\mathbb{Z}_N$ action into diagonal 2×2 blocks with eigenvalues $\pm\sigma_m$, where*

$$\sigma_m = \frac{N-1}{2} - \frac{m(N-m)}{2}. \quad (3.3)$$

The Lagrangian eigenvalues are $\lambda_m^+ = \sigma_m + (N-1)/2 = (N-1) - m(N-m)/2$ (radial) and $\lambda_m^- = m(N-m)/2$ (tangential).

Proof. The Hessian commutes with the $\mathbb{Z}_N$ action; by Schur's lemma it decomposes into 2×2 blocks per Fourier mode m , with radial entry $H_{rr}^{(m)}$, tangential entry $H_{tt}^{(m)} = -H_{rr}^{(m)}$, and off-diagonal $H_{rt}^{(m)} = 0$ (by anti-symmetry of $\sin(2\pi pm/N)$ under $p \rightarrow N-p$). The Havelock identity (Paper I, §4) gives $H_{rr}^{(m)} = (N-1)/2 - T_m = (N-1)/2 - m(N-m)/2$. Adding the Lagrange shift $-2\mu_L = (N-1)/2$ yields $\lambda_m^\pm$. $\square$

Corollary 3.3 (Constrained minimum versus unconstrained saddle). *For $N \leq 7$, the N -gon is simultaneously a constrained energy minimum on $\{L = NR^2, \mathbf{P} = 0\}$ and an unconstrained saddle point of H .*

Proof. The unconstrained Hessian has $H_{rr}^{(m)} > 0$ and $H_{tt}^{(m)} < 0$ (saddle point). The Lagrange multiplier $\mu_L = -(N-1)/(4R^2)$ shifts all eigenvalues by $+(N-1)/2$, converting the saddle into a minimum on the constraint surface. The spiral deformation (2.3) leaves the conserved surface at $O(\sigma^2)$, so $H''(0) < 0$ (Theorem 3.1) does not contradict constrained stability. $\square$

Remark 3.1 (Onsager compatibility). *The constrained minimum is the relevant concept for Lyapunov stability (Arnold's energy-Casimir method). The Onsager negative-temperature argument operates on the unconstrained landscape, where the polygon is the highest- H configuration reachable within the Thomson-stable set—not a local maximum of H , but a constrained minimum at higher energy than alternatives at the same conserved quantities.*

4 Central vortex stabilization

When a central vortex of circulation κ_0 is present at the origin, the central interaction $-\kappa_0 \sum_k \ln |z_k|$ modifies both the Lagrange multiplier and the Hessian. Two effects contribute to the critical eigenvalue:

Lagrange multiplier shift. The stationarity condition gives $\mu_L = -(N-1)/4 - \kappa_0/2$ (for $R = 1$), so the shift $-2\mu_L = (N-1)/2 + \kappa_0$ increases by $+\kappa_0$ per unit.

Hessian projection. The central-vortex Hessian at vortex k is the traceless 2×2 matrix $(\kappa_0/R^2) \begin{bmatrix} \cos 2\theta_k & \sin 2\theta_k \\ \sin 2\theta_k & -\cos 2\theta_k \end{bmatrix}$, which has eigenvalue $+\kappa_0/R^2$ for radial perturbations $(\cos \theta_k, \sin \theta_k)$ and $-\kappa_0/R^2$ for tangential. For $N \geq 8$, the critical eigenvector (mode $m = \lfloor N/2 \rfloor$) is purely radial, since $H_{rt}^{(m)} = 0$ for all m (the Hessian is block-diagonal in radial/tangential coordinates by $\mathbb{Z}_N$ equivariance; Theorem 3.2). Therefore $\langle v_{\text{crit}} | \nabla^2 H_{\text{center}} | v_{\text{crit}} \rangle = +\kappa_0$.

The total shift of the critical eigenvalue is $+2\kappa_0$ per unit κ_0 (one from the Lagrange multiplier, one from the Hessian projection). Setting the shifted eigenvalue to zero:

Proposition 4.1 (Critical central vortex strength). *For $N \geq 8$, the N -gon with central vortex is a constrained energy minimum if and only if $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$, where*

$$\kappa_{\text{crit}}(N) = \begin{cases} (N^2 - 8N + 8)/16 & N \text{ even}, \\ (N-1)(N-7)/16 & N \text{ odd}. \end{cases} \quad (4.1)$$

Proof. By Theorem 3.2, the minimum constrained Lagrangian eigenvalue at $\kappa_0 = 0$ is $\lambda_{\text{min}} = (N-1) - \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$. The central vortex shifts this eigenvalue by $+2\kappa_0$: $+\kappa_0$ from the Lagrange multiplier (since $-2\mu_L$ increases by κ_0) and $+\kappa_0$ from the central-vortex Hessian projection (since the critical eigenvector is purely radial, and the central-vortex Hessian has eigenvalue $+\kappa_0/R^2$ on radial perturbations). Setting $(N-1) - m(N-m)/2 + 2\kappa_0 = 0$ with $m = \lfloor N/2 \rfloor$ gives (4.1). $\square$

The first values are:

N	≤ 7	8	9	10	11	12	13	14
κ_{crit}	0	$\frac{1}{2}$	1	$\frac{7}{4}$	$\frac{5}{2}$	$\frac{7}{2}$	$\frac{9}{2}$	$\frac{23}{4}$

These agree with the classical Thomson dynamical stability analysis [Thomson, 1883, Havelock, 1931], confirming that energy and dynamical stability criteria are equivalent.

For $N = 6$, Proposition 4.1 gives $\kappa_{\text{crit}} = -1/4$: the hexagonal ring requires an anticyclonic central vortex ($\kappa_0/\kappa < -1/4$) for destabilisation. The critical central-vortex strength decreases monotonically with N , consistent with the Thomson–Havelock criterion.

The center-vs-ring geometry is essential: multi-ring configurations are generically less stable (Section 6).

5 The $N = 7$ bifurcation

The minimum constrained eigenvalue vanishes at $N = 7$ (mode $m = 3$), making linear stability inconclusive. We show the quartic normal-form coefficient is $\alpha_0 = 45/14 > 0$, establishing nonlinear stability.

At $N = 7$, the minimum constrained Lagrangian eigenvalue is zero (mode $m = 3$). In the language of critical phenomena (Paper I, Remark 4.2), this is the critical point of a Landau mean-field transition: the spectral gap $\Delta(N) = C_1 - f_{\max}(N)$ closes linearly as $\Delta \sim -(N-7)$ with mean-field exponent $\nu = 1$, and the bifurcation is classified by the quartic term of H along the $m = 3$ neutral direction on the constraint surface. The $\mathbb{Z}_7$ -equivariant normal form is the Landau potential $V(\varepsilon) = (\Delta/2)\varepsilon^2 + (\alpha_0/4)\varepsilon^4$, where $\varepsilon = |z|$ is the amplitude of the $m = 3$ mode and $\Delta = 0$ at $N = 7$. Numerical evaluation (tracing H along the Newton-projected constraint surface) gives $\alpha > 0$. The neutral subspace of the constrained Hessian consists of two modes: the radial Fourier modes at $m = 3$ and $m = 4 = N-3$ (together forming the 2-dimensional space of symmetry-breaking perturbations), plus the rotation mode (trivially neutral by the $\text{SO}(2)$ symmetry of H).

For both neutral modes, the *unconstrained* quartic d^4H/dt^4 (where t is the unit-eigenvector amplitude) takes the same value by $\mathbb{Z}_7$ symmetry. Each of the 21 vortex pairs contributes

$$\frac{6C^2}{A^2} - \frac{12CB^2}{A^3} + \frac{3B^4}{A^4},$$

where $A = |\Delta z|^2$, $B = 2 \operatorname{Re}(\Delta z \cdot \overline{\Delta u})$, $C = |\Delta u|^2$ are evaluated at the unit-ring N -gon ($|\Delta z|^2 = 4 \sin^2(\pi p/N)$ for separation class p). Direct summation over all 21 pairs gives

$$\left. \frac{d^4H}{dt^4} \right|_{m=3} = \frac{153}{7}. \quad (5.1)$$

The quartic sum is a polynomial in coordinates of the N -gon (whose positions are 7th roots of unity) and is $\mathbb{Z}_7$ -invariant: simultaneously shifting all vortex labels by one maps the ring to itself and maps u to a rotated copy. By Galois theory, such an invariant sum lies in $\mathbb{Q}(\zeta_7)^{\mathbb{Z}_7} = \mathbb{Q}$, so the result is necessarily a rational number. The value $153/7$ is then identified by the following argument: the 21 pairs decompose into 6 separation classes $p = 1, \dots, 6$ with 7 pairs each, and the per-pair quartic (5.1) depends on p only through $A_p = 4 \sin^2(\pi p/7)$ and the mode overlap $B_p = 2 \operatorname{Re}(\Delta z \cdot \overline{\Delta u})$, $C_p = |\Delta u|^2$. These are algebraic numbers in $\mathbb{Q}(\zeta_7)$; the Galois rationality above forces a rational result. Direct algebraic summation (exact rational arithmetic; see Code Availability) gives $153/7$. The unconstrained quartic is computed by evaluating the fourth-order circulant Taylor coefficient (the explicit per-chord sum involves products of $\csc^2$ and $\csc^4$ terms coupled across modes; see Code Availability). (The $\mathbb{Z}_7$ symmetry forces the degree-4 form on the 2-dimensional neutral subspace to be proportional to $|z|^4$, so $153/7$ holds for *any* unit vector in the $\{m=3, m=4\}$ subspace.)

The constraint-surface correction admits an exact closed form via the *log-scaling identity*. The Newton projection onto $\{L = N\}$ sends $\varepsilon \mapsto z(\varepsilon) = (z + \varepsilon u) \sqrt{N/L(\varepsilon)}$, where $L(\varepsilon) = N + \varepsilon^2$ exactly (since $\operatorname{Re} \sum_k z_k \overline{u_k} = 0$ for the radial $m=3$ mode and $\|u\|^2 = 1$). Because the logarithmic energy is a sum of $-\ln|\cdot|$ over $N(N-1)/2$ pairs, the overall scale factor $s(\varepsilon) = \sqrt{N/(N + \varepsilon^2)}$ contributes an additive term:

$$H_{\text{constrained}}(\varepsilon) = H_{\text{unconstrained}}(\varepsilon) + \frac{N(N-1)}{4} \ln \left(1 + \frac{\varepsilon^2}{N} \right). \quad (5.2)$$

Expanding the logarithm: $\ln(1 + \varepsilon^2/N) = \varepsilon^2/N - \varepsilon^4/(2N^2) + O(\varepsilon^6)$, so the 4th derivative of the correction term at $\varepsilon = 0$ is

$$\frac{N(N-1)}{4} \cdot \left(-\frac{12}{N^2} \right) = -\frac{3(N-1)}{N}.$$

For $N = 7$ this is $-18/7$. Combined with the unconstrained value $153/7$, the constrained quartic is therefore

$$\left. \frac{d^4 H}{dt^4} \right|_{\text{constrained}} = \frac{153}{7} - \frac{18}{7} = \frac{135}{7}, \quad (5.3)$$

an exact rational result (Richardson extrapolation at four step sizes confirms $135/7$ to 50 decimal places).

Kurakin and Yudovich [2002] established nonlinear stability of the heptagonal ring using Arnold's energy-Casimir method, proving positivity of the quartic form by Dirichlet's criterion without computing its value. Cabral and Schmidt [1999] and Schmidt [2004] independently proved stability via Birkhoff normal form and computed a quartic coefficient of $9/28$ in the *Sylvester-normalized* coordinates (where all quadratic terms are rescaled to $1/2$). The present value $\alpha_0 = 45/14$ is expressed in the physical (un-normalized) radial-mode coordinates on the constraint surface; both values confirm $\alpha_0 > 0$ but measure different quantities. The present derivation proceeds via the constrained energy-minimum framework (Theorem 3.2): the log-scaling identity (5.2) yields the exact correction $-3(N-1)/N$, and the unconstrained value $153/7$ is established by the rationality argument of §5 combined with direct algebraic summation. In the normal form $\mathcal{H} = (\alpha_0/4)|z|^4$, the exact constrained quartic gives

$$\alpha_0 = \frac{135/7}{6} = \frac{45}{14} \approx 3.214 > 0: \quad (5.4)$$

the energy-Casimir functional $\mathcal{H} = H + |\mu_L|L$ has a strict local minimum along the $m = 3$ direction at quartic order, $\mathcal{H} \geq \mathcal{H}_0 + (\alpha_0/4)\varepsilon^4$. The $N = 7$ Hessian is positive semi-definite (not positive definite); strict stability is established at quartic order. By the Dirichlet stability criterion [Boatto and Cabral, 2003], the heptagonal configuration is nonlinearly stable. The quartic (rather than quadratic) confinement explains why heptagonal rings are stable but more easily perturbed than hexagonal ones.

6 Multi-ring configurations and the center-ring dichotomy

Proposition 6.1 (Center-ring dichotomy). *Let N outer vortices sit at the regular N -gon on a ring of radius R_2 , and let $M \geq 2$ inner vortices sit at the regular M -gon on a ring of radius $R_1 < R_2$.*

- (i) *If $N \leq 7$ (outer ring Thomson-stable), the two-ring configuration is unstable: the Lagrangian Hessian has at least one negative eigenvalue.*
- (ii) *If $N \geq 8$ (outer ring Thomson-unstable), the inner ring can reduce but not eliminate the instability: the two-ring Hessian retains at least one negative eigenvalue.*

In contrast, a single central vortex ($M = 0$) at the $\mathbb{Z}_N$ fixed point produces a uniform eigenvalue shift $+\kappa_0/R^2$ across all Fourier modes and can achieve full stabilization (Proposition 4.1).

Proof. The proof splits into two cases.

Case 1: $N \geq 8$ (outer ring Thomson-unstable). The Havelock eigenvalue $\lambda_{\lfloor N/2 \rfloor} = (N-1) - \lfloor N/2 \rfloor(N - \lfloor N/2 \rfloor)/2 < 0$, so the outer N -ring alone has a negative constrained

eigenvalue. Adding $M \geq 2$ inner vortices introduces new degrees of freedom but does not remove the existing instability: the negative outer mode persists (its eigenvalue is a continuous function of the coupling strength, starting negative at zero coupling). This proves (ii).

Case 2: $N \leq 7$ (outer ring Thomson-stable). The two concentric regular polygons form an equilibrium at the co-rotation ratio

$$R_2/R_1 = \sqrt{(N+M-1)/(M-1)}, \quad (6.1)$$

determined by the Gauss mean-value theorem: the inner ring at $R_1 < R_2$ sees only its own M vortices, rotating at $(M-1)\kappa/(4\pi R_1^2)$ (self-interaction only), while the outer ring encloses the inner ring's total circulation $M\kappa$ and rotates at $(N+M-1)\kappa/(4\pi R_2^2)$. Equating gives (6.1).

At this equilibrium, the constrained Lagrangian Hessian $\mathcal{H}$ is computed for all 12 cases ($N \in \{5, 6, 7\}$, $M \in \{2, \dots, N-1\}$). Non-positive-definiteness is certified by two algebraic tests:

- **Determinant test** (8 cases with odd number of negative eigenvalues): $\det(\mathcal{H}) < 0$ implies an odd number of negative eigenvalues, hence ≥ 1 .
- **Sylvester minor test** (4 remaining cases with even number of negative eigenvalues): the k -th leading principal minor of $\mathcal{H}$ is negative for some $k \leq \dim(\mathcal{H})$, so $\mathcal{H}$ is not positive semi-definite.

N	M	$\lambda_{\min}$	n_{neg}	Certificate
3	2	-6.148	2	minor(3) < 0
4	2	-4.131	2	minor(5) < 0
4	3	-7.872	3	det < 0
5	2	-0.375	3	det < 0
5	3	-1.410	3	det < 0
5	4	-3.085	4	minor(7) < 0
6	2	-0.204	2	minor(8) < 0
6	3	-0.795	3	det < 0
6	4	-1.688	4	minor(9) < 0
6	5	-3.136	5	det < 0
7	2	-0.088	1	det < 0
7	3	-0.316	3	det < 0
7	4	-0.803	3	det < 0
7	5	-1.691	5	det < 0
7	6	-3.012	6	minor(9) < 0

Both tests are verified in exact rational arithmetic (see Code Availability).

Complete proof. All fifteen Thomson-stable cases ($N \leq 7$, $2 \leq M \leq N-1$) are verified by exact rational arithmetic (certificates in Supplement S, §2). For $M \geq 8$: the inner ring's own Thomson instability ($N_{\text{crit}} = 7$) produces a negative eigenvalue, and the inter-ring tangential Hessian $\cos(2\theta)/d^2$ is negative semidefinite in the Fourier basis when the outer ring is exterior (Supplement S, §2), so the coupling only deepens the instability. $\square$

Jupiter's observed center-plus-octagon geometry is therefore *stabilized within the center/ring configuration family*: among all configurations with a secondary vortex ring, only the single central cyclone achieves full stabilization of the $N = 8$ ring.

7 From point vortices to continuous vorticity

7.1 The vortex blob bridge (Jupiter regime)

The results of the preceding sections are stated for point vortices. The RG classification (Paper I, Proposition 2.1) guarantees that compact vorticity distributions interact logarithmically at large separation, with irrelevant corrections $O(\varepsilon^2/d^2)$. Two distinct bridges make this quantitative.

For discrete cyclone configurations ($\beta R^3/\kappa \ll 1$), replace each point vortex with a smooth blob of width ε (blob width throughout this section; unrelated to the quartic perturbation amplitude of §5). The regularized interaction $h_\varepsilon(d) = -\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$ recovers $-\ln d$ as $\varepsilon \rightarrow 0$.

Proposition 7.1 (Blob stability correction). *The constrained Lagrangian eigenvalues for h_ε satisfy*

$$\lambda_m(\varepsilon) = \lambda_m(0) + \left[-\frac{N^2-1}{12} + P_m \right] \varepsilon^2 + O(\varepsilon^4), \quad (7.1)$$

where $\lambda_m(0)$ is the Havelock eigenvalue (Theorem 3.2).

Proof. Lagrange multiplier shift. The stationarity condition $\partial H_\varepsilon / \partial x_0 = \mu_L \cdot 2x_0$ gives $\mu_L(\varepsilon) = \mu_L(0) + \delta\mu_L \cdot \varepsilon^2$ where $\delta\mu_L = (N^2-1)/24$, using $\sum_{k=1}^{N-1} \csc^2(\pi k/N) = (N^2-1)/3$. The shift $-2\delta\mu_L = -(N^2-1)/12$ is *universal* (the same for all Fourier modes).

Hessian correction. The $O(\varepsilon^2)$ corrections to h'' and h'/d are: $\delta(h''(d)) = -6\varepsilon^2/d^4 + O(\varepsilon^4)$, $\delta(h'(d)/d) = 2\varepsilon^2/d^4 + O(\varepsilon^4)$. For a pair at distance $d_p = 2 \sin(\pi p/N)$, the Hessian correction per pair is $O(\varepsilon^2/d_p^4)$. The projection $P_m = v_m^\top \delta \nabla^2 H v_m$ of this correction onto the critical eigenvector v_m is computed by summing over all $N-1$ pairwise distances with the Fourier mode weights.

Operator norm bound. Since $d_{\min} = 2 \sin(\pi/N) > 0$:

$$\|\nabla^2 H_\varepsilon - \nabla^2 H_0\|_{\text{op}} \leq 6(2N-3) \varepsilon^2 / d_{\min}^4 = O(\varepsilon^2). \quad (7.2)$$

By Weyl's perturbation theorem, each eigenvalue shifts by at most this amount, giving uniform convergence at $O(\varepsilon^2)$.

Profile independence. For a blob with finite second moment σ^2 (Gaussian, compact support), the Fourier transform $|\hat{\phi}(\varepsilon k)|^2 = 1 - \sigma^2 \varepsilon^2 |k|^2 + O(\varepsilon^4 |k|^4)$ shows the $O(\varepsilon^2)$ correction is a constant in k -space (= self-energy, not affecting pairwise eigenvalues). The pairwise correction is $O(\varepsilon^4)$ —*faster* than the Cauchy blob. The Cauchy blob has algebraic tails ($\sim r^{-2}$) and infinite second moment ($\sigma^2 = \infty$), so its Fourier transform $\hat{\phi}(\varepsilon k) = e^{-\varepsilon|k|}$ decays at rate ε rather than ε^2 . Among profiles with finite σ^2 (Gaussian: $e^{-\sigma^2 k^2/2}$; compact support: $\hat{\phi} = 0$ for $|k| > 1/\varepsilon$), the pairwise correction is $O(\varepsilon^4)$ —strictly faster than the Cauchy $O(\varepsilon^2)$. The Cauchy blob is therefore the slowest-converging profile, and all finite- σ^2 profiles inherit the stability conclusions at $O(\varepsilon^2)$ or better. $\square$

The Hessian projection P_m and the total correction $c_m = -(N^2-1)/12 + P_m$ (specific to the Cauchy blob $h_\varepsilon = -\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$) are exact rational numbers, computed as follows. Differentiating the Cauchy blob pair Hessian with respect to ε^2 at $\varepsilon = 0$ gives the per-pair correction:

$$\delta H_{jj,xx} = \frac{2}{d_{jk}^6} \left((\Delta y)^2 - 3(\Delta x)^2 \right), \quad \delta H_{jk,xx} = -\delta H_{jj,xx}, \quad (7.3)$$

with analogous expressions for yy and xy (see proof of Proposition 7.1). Here $(\Delta x, \Delta y, d_{jk})$ are the pair separation components. The critical radial Fourier mode is $v_k \propto z_k \cos(2\pi mk/N)$ with $m = \lfloor N/2 \rfloor$. The projection

$$P_m = v_m^\top \delta \nabla^2 H v_m \quad (7.4)$$

is a finite quadratic form: a sum over $N(N-1)$ pair contributions, each involving trigonometric values at N -th roots of unity. Since $\sin(\pi p/N)$ and $\cos(2\pi p/N)$ are algebraic for integer p and N , every term is rational and so is the sum. Explicit values (verified to machine precision; see Code Availability):

N	3	4	5	6	7	8
$-(N^2-1)/12$	$-\frac{2}{3}$	$-\frac{5}{4}$	-2	$-\frac{35}{12}$	-4	$-\frac{21}{4}$
P_m	$-\frac{1}{3}$	$+\frac{1}{4}$	$+1$	$+4$	$+8$	$+\frac{65}{4}$
c_m	-1	-1	-1	$+\frac{13}{12}$	$+4$	$+11$

The Hessian projection P_m changes sign between $N = 3$ ($P_3 = -1/3$) and $N = 4$ ($P_4 = +1/4$): for the triangle, the blob softening reduces the nearest-neighbor curvature (all pairs are at the same short distance $d = \sqrt{3}$), while for the square and larger polygons, the long-range pairs increasingly dominate the projection and $P_m > 0$. Despite this sign change, $c_m = -1$ universally for $N \leq 5$ because the Lagrange multiplier shift $-(N^2-1)/12$ dominates P_m in this range. For $N \geq 6$, P_m grows faster and overwhelms the shift.

By profile independence (proved above), all finite- σ^2 blobs have smaller corrections, so the stability conclusions hold *a fortiori*. Three consequences follow:

(i) *Stability preserved.* For $N \leq 5$: $\lambda_0 \geq 1$ and $c = -1$, so $\lambda(\varepsilon) = \lambda_0 - \varepsilon^2 > 0$ for $\varepsilon < 1$. For $N = 6$: $c > 0$, so $\lambda(\varepsilon) > \lambda_0$ —the blob *enhances* stability.

(ii) *Marginal mode resolved.* At $N = 7$, $\lambda_0 = 0$ and $c = +4$, so $\lambda(\varepsilon) = 4\varepsilon^2 > 0$ for any $\varepsilon > 0$. The heptagonal marginal stability is an artifact of the singular point vortex limit.

(iii) *Instability partially compensated.* For $N = 8$: $\lambda_0 = -1$ and $c = 11$, so the octagon is stabilized for $\varepsilon > 1/\sqrt{11} \approx 0.30$.

7.1.1 Full infinite-dimensional convergence

The full infinite-dimensional stability analysis (including blob shape deformations and center-shape coupling) is given in Appendix A. The main result (Proposition A.1) is that for sufficiently concentrated blobs, the full Hessian inertia equals the point vortex Havelock inertia: no eigenvalue is lost, no sign is changed, and no new instability arises from the infinite-dimensional shape degrees of freedom. Corollaries A.2 and A.3 give the explicit concentration bounds for $N \leq 6$ and the marginal-mode resolution for $N = 7$.

7.2 The Rossby bridge (Saturn regime)

For continuous jet meanders ($\beta R^3/\kappa \gg 1$), the bridge is not vortex concentration but *Rossby wave quantization*. The QGPV linearized about a zonal jet $U(y)$ gives the Rayleigh–Kuo equation:

$$(U - c)(\hat{\psi}'' - k^2 \hat{\psi}) + (\beta - U'')\hat{\psi} = 0.$$

For a stationary wave ($c = 0$) at the jet center ($y = 0$) of a Gaussian jet with $U''(0) = -U_{\max}/\sigma^2$:

$$\hat{\psi}'' + \left[\frac{\beta_{\text{eff}}}{U_{\max}} - \frac{n^2}{R^2} \right] \hat{\psi} = 0, \quad \beta_{\text{eff}} = \beta + \frac{U_{\max}}{\sigma^2},$$

where the effective PV gradient β_{eff} includes the jet curvature (which dominates β by a factor ~ 13 for Saturn). The stationary azimuthal wavenumber is

$$n^* = R_S \cos \varphi \sqrt{\beta_{\text{eff}}/U_{\max}}, \quad (7.5)$$

where $R_S \cos \varphi$ is the ring radius at latitude φ ($\approx 1.31 \times 10^7$ m for Saturn at 76° N). The nearest integer $n = \text{round}(n^*)$ is the selected polygon wavenumber. For Saturn: $\beta_{\text{eff}}/U_{\max} \approx 1.72 \times 10^{-13} \text{ m}^{-2}$, giving $n^* \approx 5.4$ from the leading-order dispersion relation. The observed $n = 6$ is obtained in numerical eigenvalue computations of the full Rayleigh–Kuo equation [Sánchez-Lavega et al., 2014]; the discrepancy between the leading-order $n^* \approx 5.4$ and the observed $n = 6$ arises from the assumed jet width: equation (7.5) uses $\sigma \approx 2000$ km (Gaussian fit to the Cassini profile), but the observed half-width is 1500–2500 km. At $\sigma \approx 1750$ km (15% narrower, well within measurement uncertainty), $\beta_{\text{eff}}/U_{\max}$ increases by 22% and $n^* = 6.0$ exactly. The sensitivity: $\partial \ln n^* / \partial \ln \sigma = -1$ (one unit of jet narrowing gives one unit of wavenumber increase).

The discretization mechanism is *azimuthal periodicity*: the wave must fit an integer number of wavelengths around the ring.

The connection to the point vortex energy landscape proceeds through the *critical layer* of the saturated wave. At finite amplitude, the stationary Rossby wave develops a cat’s-eye structure at the critical line $U = c = 0$: vorticity is trapped into N lobes of width $\sim \sigma$ centered at the meander peaks. As the jet sharpens ($\sigma/R \rightarrow 0$ at fixed wave amplitude), these lobes concentrate into N isolated patches near the N -gon vertices.

Remark 7.1 (Cat’s-eye Hessian convergence). *The cat’s-eye vorticity decomposes as $\omega = \omega_{\text{lobe}} + \omega_{\text{braid}} + \omega_{\text{bg}}$, where the lobes are the trapped fluid inside the separatrices, the braids are thin filaments along the separatrix itself, and the background is the passing fluid outside the cat’s-eye.*

Two conditions are required for Proposition A.1 to apply:

(i) **Lobe concentration.** *The lobes satisfy the concentration condition (A.2) with $\varepsilon \sim \sigma$ and the centroids converge to the N -gon vertices. This is guaranteed by $\mathbb{Z}_N$ symmetry.*

(ii) **Braid smallness.** *The absolute braid circulation Γ_{braid} is negligible compared to the total lobe circulation Γ_{lobe} .*

Both conditions hold for the Stuart vortex $\Psi = -\sigma^2 \ln(\cosh(y/\sigma) - \eta \cos(n\theta))$ ($\eta \in (0, 1)$; $\eta \rightarrow 1$ is point-vortex concentration).

Proof of lobe dominance. The Stuart vortex satisfies $\omega = C/F^2$ with $F = \cosh(y/\sigma) - \eta \cos(n\theta)$. Near the elliptic center ($\theta = 0, y = 0$): $F \approx (1-\eta) + (y/\sigma)^2/2 + n^2\eta\theta^2/2$, so

$$\Gamma_{\text{lobe}} = C \iint_{\text{lobe}} \frac{dA}{F^2} \geq C \int_0^\infty \frac{2\pi r dr}{[(1-\eta) + r^2/2]^2} = \frac{2\pi C}{1-\eta}.$$

Outside the separatrix, $F \geq 1+\eta \geq 1$, so $\Gamma_{\text{non-lobe}} = C \iint_{\text{outside}} F^{-2} dA \leq C A_{\text{period}}$, which is bounded independently of η . Therefore

$$\frac{\Gamma_{\text{non-lobe}}}{\Gamma_{\text{total}}} \leq \frac{A_{\text{period}}(1-\eta)}{2\pi + A_{\text{period}}(1-\eta)} = O(1-\eta). \quad (7.6)$$

The lobe fraction $\Gamma_{\text{lobe}}/\Gamma_{\text{total}} = 1 - O(1-\eta)$ as $\eta \rightarrow 1$. Numerically: at $\eta = 0.99$, lobes capture $> 99\%$ of the total circulation.

Combining with Proposition A.1: the cat’s-eye satisfies the blob bridge conditions when $\sigma/d_{\min} < \delta(N)/\sqrt{2}$ and $1-\eta < c\delta(N)^2$. The first condition is the standard concentration bound on the lobe width. The second condition holds because the braid perturbation is $O(1-\eta)$ of the lobe system (equation (7.6)). Both conditions are satisfied for η sufficiently close to 1 (the point-vortex limit). Therefore, by Proposition A.1, the constrained Hessian inertia of the cat’s-eye equals the Havelock inertia. The numerical verification ($\eta \in [0.3, 0.99]$; see Code Availability) confirms this.

The same argument applies to the Prandtl–Batchelor cat’s-eye, the physically relevant model for Saturn’s hexagon, where the lobe dominance is exact: the PB theorem [Batchelor, 1956] gives $\omega = \omega_{\text{lobe}}$ (constant) inside each separatrix and $\omega = \omega_{\text{bg}}$ outside. The braid (the separatrix curve $\psi = \psi_{\text{sep}}$) is a single streamline of zero area, so $\Gamma_{\text{braid}} = \int_{\text{separatrix}} \omega dA = 0$ exactly. Every unit of circulation is in either lobe or background; lobe dominance (7.6) holds with $1-\eta$ replaced by 0. See Code Availability.

The Saturn bridge thus arrives at the *same* energy landscape as the Jupiter bridge, through a different limiting procedure: Saturn concentrates a continuous jet meander into N patches (jet width $\sigma \rightarrow 0$), while Jupiter concentrates N discrete cyclones into point vortices (blob radius $\varepsilon \rightarrow 0$). Both limits converge to the Havelock spectrum.

8 Finite Rossby deformation radius: the K_0 interaction

The K_0 interaction provides a second example of the RG flow toward the logarithmic fixed point (Paper I, Proposition 2.1): the exponential tail $K_0(r/R_d) + \ln r = O(e^{-r/R_d})$ is an irrelevant perturbation (faster than any power law), and the limit $R_d \rightarrow \infty$ is the flow to the fixed point.

In quasi-geostrophic dynamics with finite Rossby deformation radius R_d , the Green’s function is $K_0(r/R_d)$ (modified Bessel function), which recovers $-\ln r$ for $r \ll R_d$. The interaction $h(r) = K_0(r/R_d)$ satisfies $h'(r) = -K_1(r/R_d)/R_d < 0$ and $(rh'(r))' = rK_0(r/R_d)/R_d^2 > 0$ (class (iii)). Numerical computation of the constrained Lagrangian Hessian as a function of $\eta = R/R_d$ (polygon ring radius over deformation radius) reveals an N -dependent *polygon-suppression transition*:

N	Barotropic $\lambda_{\min}$	Critical R/R_d	Status
≤ 5	≥ 1	none	Stable for all R/R_d
6	$1/2$	≈ 1.29	Suppressed for $R/R_d > 1.29$
7	0	≈ 0	Unstable for any finite R/R_d
≥ 8	< 0	N/A	Unstable (barotropic)

Polygons with $N \leq 5$ are robust against stratification effects. The hexagon ($N = 6$) is the first polygon sensitive to R_d : it requires $R/R_d < 1.29$. The heptagon ($N = 7$), neutrally stable in the barotropic limit, is destabilized by any finite R/R_d —providing an additional mechanism for its rarity beyond the quartic confinement.

For Saturn, the barotropic deformation radius ($\sim 2000\text{--}3000$ km) gives $R/R_d \approx 8\text{--}12$ at the hexagonal jet, placing it above the critical ratio. The observed hexagonal persistence therefore requires the dynamics to be effectively barotropic over a deep layer—a

requirement now directly supported by observational data. Cassini Grand Finale gravity measurements constrain Saturn’s differential rotation to a depth of 8,000–9,000 km [Iess et al., 2019], implying a coherent barotropic column of order 10^4 km, far exceeding the hexagon’s vertical scale. Saturn is therefore the best-supported case in our sample for the applicability of the 2D logarithmic Green’s function: the columnar depth is observationally established.

Observable prediction: hexagon suppression signatures. The critical ratio $R/R_d \approx 1.29$ for $N = 6$ yields a testable prediction. If the effective barotropic depth were reduced (by seasonal radiative forcing of the upper troposphere, or by a hypothetical change in deep convective coupling), R_d would decrease and R/R_d would increase toward the critical value. Approaching the threshold, the framework predicts three observable signatures: (1) the minimum Havelock eigenvalue $\lambda_{\min}(\eta) \rightarrow 0$, reducing the hexagonal mode’s restoring frequency and producing a *longer oscillation period* for meridional meander perturbations; (2) the hexagonal amplitude ε_{obs} would *grow*, since the restoring force weakens while convective forcing remains constant; and (3) at $R/R_d > 1.29$, the hexagonal mode becomes linearly unstable and the pattern would transition to a lower- N polygon ($N = 5$, the next Thomson-stable configuration) or break up entirely. Saturn’s hexagon has been observed continuously since Voyager 1 (1980–81); Cassini monitored it through northern spring equinox (2009) to summer solstice (2017) with no sign of weakening [Antuñano et al., 2015, Sánchez-Lavega et al., 2014]. This four-decade persistence across half a Saturnian year places an empirical lower bound on the effective barotropic depth: seasonal radiative forcing does not reduce R_d enough to approach the critical ratio, consistent with the deep ($\sim 10^4$ km) barotropic column implied by Cassini gravity.

Why the leading K_0 correction does not shift stability. The small- r expansion $K_0(r/R_d) = -\ln(r/R_d) - \gamma_E + r^2/(4R_d^2) + O(r^4)$ shows the leading correction to the logarithm is $g(r) = r^2/(4R_d^2)$ —a perturbation of RG exponent $\alpha = 2$. By the isotropic-marginality result (Paper I, Proposition 2.2), the pair Hessian anisotropy $g'' - g'/r = \alpha(\alpha-2)r^{\alpha-2}$ vanishes at $\alpha = 2$: the Hessian perturbation is purely isotropic ($\propto I$) and is exactly compensated by the rotation-frequency shift $\delta\Omega$. Consequently $\delta\lambda_m = 0$ for all binding modes—the $r^2/(4R_d^2)$ term shifts the precession rate of the polygon but does *not* shift its stability eigenvalues. The first correction to stability from K_0 is $O(r^4/R_d^4)$ ($\alpha = 4$, which is *not* isotropic-marginal), and this is numerically small for $R/R_d \ll 1$. This explains why the barotropic ($R_d \rightarrow \infty$) stability predictions remain accurate even at moderate R/R_d : the leading deformation-radius correction is stability-neutral.

Jupiter polar cyclones. The situation is different, and the 2D justification requires separate examination. Juno gravity measurements [Kaspi et al., 2018] constrain Jupiter’s differential rotation to $\sim 3,000$ km depth ($\approx 4\%$ of R_J), far shallower than Saturn’s. Jupiter’s polar cyclones therefore do *not* rest on deep barotropic Taylor columns; the relevant 2D dynamics is that of a shallow weather layer above a near-solid-body rotating deep interior. Two limiting cases bracket the problem:

Deep weather-layer limit ($H_{\text{eff}} \sim 3,000$ km). Taking the full differential-rotation depth as the effective layer and using the equivalent-barotropic deformation radius $R_d = \sqrt{g H_{\text{eff}}/f} \approx \sqrt{25 \times 3 \times 10^6 / (2\Omega_J \sin 84^\circ)} \approx 25,000$ km, we obtain $R/R_d \approx 0.3 \ll 1$. In this limit $K_0(r/R_d) \rightarrow -\ln r + \text{const}$, so the logarithmic Green’s function applies—

though the physical justification is a shallow equivalent-barotropic layer rather than a deep Taylor column.

Tropospheric limit ($H_{\text{eff}} \sim 30\text{--}100\text{ km}$). If the active polar-cyclone layer is shallower, reduced gravity $g' \sim 0.1\text{--}1\text{ m s}^{-2}$ gives $R_d \sim 2,000\text{--}6,000\text{ km}$ and $R/R_d \sim 1\text{--}4$. The $N = 8$ ring lies in the finite- R_d regime. Computing the κ_0 -shifted constrained Hessian eigenvalue for the K_0 interaction with $\kappa_0/\kappa = 0.7$ (the Juno central estimate):

N	κ_0/κ	$(R/R_d)_{\text{crit}}$	Tropospheric stable?
6	0.7	17.2	yes ($\gg 4$)
7	0.7	4.2	marginal
8	0.7	0.79	no (< 1)
9	0.7	0.10	no

Here $(R/R_d)_{\text{crit}}$ is the ratio at which the minimum constrained eigenvalue crosses zero: the ring is stable for $R/R_d < (R/R_d)_{\text{crit}}$ and unstable above. For $N = 8$ the critical ratio is $(R/R_d)_{\text{crit}} \approx 0.79$, which lies *below* the tropospheric range $R/R_d \in [1, 4]$. **The $N = 8$ prediction therefore requires the deep weather-layer limit** ($R/R_d \lesssim 0.8$, i.e. $R_d \gtrsim 1.3 R \approx 9,700\text{ km}$); it does not survive into the tropospheric regime.

The effective layer depth is not directly constrained by current Juno data, making Jupiter the *least* observationally supported case for the 2D assumption, in contrast to Saturn. A direct measurement of the depth of the polar cyclone roots—analogueous to the gravity constraints on the jets—would resolve this ambiguity. The computation above sets a quantitative target: $R_d > 9,700\text{ km}$ is necessary for $N = 8$ stability under the K_0 interaction.

The critical ratio is defined implicitly by the equation $\bar{\lambda}(\eta_{\text{crit}}, N) = 0$, where the dimensionless eigenvalue $\bar{\lambda}(\eta, N) = R^2 \cdot \lambda_{\text{min}}(\eta)$ interpolates smoothly from the Havelock value $\bar{\lambda}(0, N) = (N-1) - \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$ to zero.

The Lagrange multiplier is determined by the stationarity condition $\partial H/\partial R + \mu_L \partial L/\partial R = 0$. For $H = -\sum_{j < k} K_0(d_{jk}/R_d)$ with $d_{jk} = 2R \sin(\pi p/N)$ and $L = NR^2$, the radial derivative of each pair contributes $-K'_0(d_{jk}/R_d) \cdot 2 \sin(\pi p/N)/R_d = K_1(d_{jk}/R_d) \cdot 2 \sin(\pi p/N)/R_d$ (using $K'_0 = -K_1$). Projecting onto the constraint gradient $\partial L/\partial R = 2NR$ and solving gives

$$-2\mu_L = \frac{\eta}{R} \sum_{p=1}^{N-1} \frac{(1 - \cos \theta_p) K_1(2\eta \sin(\pi p/N))}{2 \sin(\pi p/N)}, \quad \theta_p = 2\pi p/N, \quad (8.1)$$

where $\eta = R/R_d$ and the factor $(1 - \cos \theta_p)$ arises from the Fourier decomposition of the pairwise Hessian along the radial mode. As $\eta \rightarrow 0$: $K_1(x) \sim 1/x$ for small x , so each summand reduces to $(1 - \cos \theta_p)/(4\eta \sin^2(\pi p/N)) \cdot (\eta/R) = (1 - \cos \theta_p)/(4R \sin^2(\pi p/N))$. Summing over p using the Havelock identity $\sum_p (1 - \cos \theta_p)/(4 \sin^2(\pi p/N)) = (N-1)/2$ recovers the flat-plane value $-2\mu_L = (N-1)/(2R^2)$. For $N = 6$, the critical mode $m = 3$ has zero Havelock weight at $p = 2, 4$ (since $\cos(2\pi \cdot 2 \cdot 3/6) = 1$), so the critical eigenvalue depends on K_0 and K_1 at only *two* distinct arguments, η and 2η , rather than three.

No closed-form analog of the Havelock identity is known for K_0 . The orthogonal-ity mechanism that collapses the logarithmic sum to $m(N-m)/2$ relies on the identity $\sum_p e^{2\pi i m p/N} \ln |2 \sin(\pi p/N)| = 0$, which has no counterpart for K_0 : the Bessel values $K_0(2\eta \sin(\pi p/N))$ are transcendental at nonzero algebraic arguments by Siegel's theorem on E -functions, but this does not preclude the full weighted sum from having a closed form. Whether $\sum_p (\cos \theta_{pm}) K_0(2\eta \sin(\pi p/N))$ admits a closed form for any N and m is

an open question; no such form has been found, and all stability computations for K_0 interactions are carried out numerically. A perturbative expansion $K_0(x) \approx -\ln x + O(x^2)$ gives $\eta_{\text{crit}} \approx 0.9$ for $N = 6$ —30% below the exact value—because the critical ratio lies in the non-perturbative regime where K_0 departs significantly from $-\ln$. The transition occurs precisely where the exponential decay of K_0 (destroying long-range correlations between non-adjacent vortices) overtakes the logarithmic regime, making the suppression inherently non-perturbative. The implicit equation is nevertheless well-posed, and the critical η is uniquely determined by bisection for each N .

The two phase boundaries. For K_0 , two distinct boundaries exist in the (N, η) plane: (1) the *spiral sign change* $\bar{H}''(0) = 0$, where the positive A_m term from $(rh')' > 0$ overwhelms the negative B_m term; and (2) the *constrained stability loss* $\bar{\lambda}_{\min} = 0$, where the Lagrange multiplier shift consumes the stability margin. The constrained boundary ($\eta \approx 1.29$ for $N = 6$) is always *inside* the spiral boundary ($\eta \approx 3.2$ for $N = 6$) because the Lagrange shift reduces the margin before H'' changes sign. Physically, the polygon requires long-range correlations between non-adjacent vortices. When $\eta \gg 1$ and the K_0 interaction decays exponentially, only nearest-neighbor correlations survive and the polygon loses its energetic advantage.

For comparison, the Cauchy-blob regularization $-\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$ has $H'' < 0$ for *all* ε —the soft-core never flips the spiral sign. The Gaussian-screened interaction $-\ln(d) \cdot e^{-d^2/L^2}$ shows an earlier transition ($\eta \approx 0.9$ for $N = 6$) due to its faster-than-exponential decay.

The constrained stability boundary. The interaction class determines N_{crit} , defined here as the largest N for which the *constrained* Lagrangian Hessian $\bar{\lambda}_{\min} \geq 0$ (i.e. the polygon is a constrained energy minimum). This is distinct from the spiral-direction boundary $H''(0) = 0$, which occurs at larger η (e.g. $\eta \approx 3.2$ vs. 1.29 for $N = 6$ with K_0 ; see the two phase boundaries above):

Class	Condition	N_{crit} (constrained)
(i)	$(rh')' < 0$	≥ 7 (increases with $ (rh')' $)
(ii)	$(rh')' = 0$ (only $-\ln r$)	$= 7$ exactly
(iii)	$(rh')' > 0$ (e.g. K_0)	≤ 7 (decreases with screening)

Numerical computation confirms: for $h = -r^\alpha$ (class (i)), $N_{\text{crit}} = 7$ at $\alpha \rightarrow 0$ and increases with α ($N_{\text{crit}} = 8$ at $\alpha = 1$, $N_{\text{crit}} = 14$ at $\alpha = 2$). At $N = 7$, $\lambda_{\min}(\alpha) > 0$ for all $\alpha > 0$ tested, with $d^2\lambda_{\min}/d\alpha^2 = +2.5 > 0$ at $\alpha \rightarrow 0$: the marginal stability at $N = 7$ is *unique to the logarithmic interaction*, and any class (i) interaction makes the heptagon strictly stable. For $K_0(r/R_d)$ (class (iii)), the dimensionless eigenvalue $\bar{\lambda}(\eta)$ decreases monotonically in η for each Fourier mode (verified for $N = 5, 6$), consistent with the physical expectation that screening weakens the stabilizing long-range correlations. For the K_0 interaction, $N_{\text{crit}} = 6$ at $R/R_d = 1$ and decreases to 5 at $R/R_d = 3$. The logarithmic interaction is the *watershed*: the critical case between interactions that stabilize polygons (class (i)) and interactions that suppress them (class (iii)).

Proposition 8.1 (Min/max duality). *For any pairwise interaction $h(r)$, the constrained energy minimum of $H = \sum h(d_{jk})$ and the constrained energy maximum of $-H$ have the same stability boundary N_{crit} .*

Proof. The Lagrange multiplier for $-H$ is $\mu_L^{-H} = -\mu_L^H$ (from $-\nabla H = (-\mu_L)\nabla L$). The Lagrangian Hessian $\mathcal{L}_{-H} = -\nabla^2 H + 2\mu_L I = -\mathcal{L}_H$. The eigenvalues are negated; constrained minimum (all ≥ 0) of H is equivalent to constrained maximum (all ≤ 0) of $-H$. $\square$

For gravity ($h = -1/r$, $h' > 0$), the N -gon is a constrained energy maximum. The duality gives $N_{\text{crit}}^{\text{gravity}} = N_{\text{crit}}^{\text{Coulomb}}$. Numerical computation of the Coulomb ($h = +1/r$) constrained Hessian gives $N_{\text{crit}} = 5$: the minimum eigenvalue is $+0.73$ at $N = 5$ and -0.02 at $N = 6$. This is consistent with Maxwell’s (1859) result that gravitating rings require a central mass for $N \geq 7$; the one-unit difference ($N_{\text{crit}} = 5$ vs Maxwell’s 6) arises because our analysis does not include the central mass (Maxwell’s stability includes the stabilizing effect of the central body).

Maxwell (1859) obtained $N_{\text{crit}} = 6$ for the Coulomb ring *with* a central mass; without the central mass, $N_{\text{crit}} = 5$. The central mass adds a uniform eigenvalue shift $+\kappa_0/R^2$ that stabilises $N = 6$.

The Coulomb/gravity $N_{\text{crit}} = 5$ is less than the logarithmic $N_{\text{crit}} = 7$, confirming the class (iii) reduction: the $1/r$ interaction has $(rh')' = 1/r^2 > 0$, placing it in the destabilizing regime relative to $-\ln r$.

9 Oblateness and the ice giants

9.1 Saturn’s oblateness determines hexagon existence

The preceding sections analysed stability on constant-curvature surfaces and with finite deformation radius. A third physical correction—planetary oblateness—has a qualitatively different effect: it can reverse the sign of an eigenvalue.

The Rossby analysis of §7.2 treats Saturn as a sphere. Saturn is an oblate spheroid with flattening $f = (a-b)/a = 0.098$ ($a = 60,268$ km equatorial, $b = 54,364$ km polar). The oblateness changes the *sign* of the Thomson stability eigenvalue for $N = 6$: it is not a small correction but the mechanism that determines whether the hexagon can exist.

The Gaussian curvature at geodetic latitude φ is $K(\varphi) = 1/(M(\varphi)N(\varphi))$ where $M = a(1-e^2)/(1-e^2\sin^2\varphi)^{3/2}$ and $N = a/(1-e^2\sin^2\varphi)^{1/2}$. On an oblate body, K is largest at the equator and smallest at the pole—the pole is flatter. For Saturn: $K_{\text{eq}}/K_{\text{pole}} = a^4/b^4 = 1.51$. Since positive curvature destabilises (Paper I, curvature dichotomy), the pole is the most stable point on the spheroid.

Substituting the local curvature into the spherical C_1 formula:

Model	$\lambda_3(N=6)$	Stable?
Mean sphere ($R_{\text{mean}} = 58,300$ km)	-0.12	No
Oblate spheroid at 78°N	$+0.01$	Marginally yes

The oblateness flips λ_3 from -0.12 to $+0.01$. On a mean sphere, the hexagon is Thomson-unstable and should not exist; on the oblate spheroid, it is marginally stable.

Proposition 9.1 (Oblateness-stabilised hexagon). *On an oblate spheroid with flattening f , the binding eigenvalue $\lambda_3(N=6)$ at the pole exceeds its mean-sphere value by*

$$\delta\lambda_3 \approx (N-1) \cdot \frac{2\xi f (2-f)}{(1+\xi)^2} \quad (9.1)$$

at leading order in f , where $\xi = K_{\text{mean}} R^2$. For Saturn ($f = 0.098$, $\xi = K_{\text{mean}} R^2 \approx (15,000/58,300)^2 = 0.066$): the leading-order formula gives $\delta\lambda_3 \approx 0.11$; the full numerical computation (Code Availability) gives $\delta\lambda_3 = 0.13$, with the difference arising from higher-order terms in f and ξ . This reverses the sign from -0.12 to $+0.01$.

Proof sketch. The Gaussian curvature at the pole of an oblate spheroid is $K_{\text{pole}} = b^2/a^4 = K_{\text{mean}}(1-f)^3$ (since $K_{\text{mean}} \approx 1/(ab)$ and $b/a = 1-f$). The sphere threshold formula $C_1^{S^2}(\xi) = (N-1)(1-\xi)/(1+\xi)$ evaluated at $\xi_{\text{pole}} = K_{\text{pole}} R^2$ gives, relative to ξ_{mean} : $\delta C_1 = -(N-1) \times 2 \delta\xi / (1+\xi)^2 + O(\delta\xi^2)$, where $\delta\xi = \xi_{\text{pole}} - \xi_{\text{mean}} = \xi_{\text{mean}}[(1-f)^3 - 1] \approx -3f \xi_{\text{mean}}$. Since the pole is flatter ($K_{\text{pole}} < K_{\text{eq}}$, hence $\delta\xi < 0$), $\delta C_1 > 0$: the eigenvalue increases. The leading-order approximation gives $\delta\lambda_3 \approx (N-1) \times 6f\xi/(1+\xi)^2$; the full formula (9.1) uses the exact $(1-f)^3$ expansion rather than the linearisation, producing the $f(2-f)$ factor. $\square$

The critical latitude φ_{crit} where $\lambda_3 = 0$ is $\approx 72^\circ\text{N}$ for Saturn. The hexagon at 78°N is 6° above this boundary—comfortably in the stable region.

At non-polar latitudes, the curvature gradient $\delta K/K \approx 4.2\%$ across the hexagon ring couples Fourier modes m and $m \pm 2$. The coupling amplitude (~ 0.19) exceeds the stability margin ($|\lambda_3| = 0.01$) by a factor of 19. By the topological protection of the Morse index (Paper I), this mode coupling cannot change the net spectral flow—the hexagon remains stable in the topological sense.

9.2 Neptune predictions: packing-dominated

Neptune ($R = 24,600 \text{ km}$, $\Omega = 1.08 \times 10^{-4} \text{ rad s}^{-1}$, $f = 0.017$) is in the vortex-dominated regime: at 84°N , $\beta_N = 2\Omega_N \cos(84^\circ)/R_N \approx 9.2 \times 10^{-13} \text{ m}^{-1}\text{s}^{-1}$ and $R_{\text{ring}} \approx 2,570 \text{ km}$, giving $\beta R^3 \approx 1.6 \times 10^7 \text{ m}^2 \text{ s}^{-1}$. For Jovian-scaled cyclones ($V_{\text{max}} \approx 80\text{--}100 \text{ m s}^{-1}$, $r_c \approx 240 \text{ km}$), $\kappa \approx 1.2\text{--}1.5 \times 10^8 \text{ m}^2 \text{ s}^{-1}$, so $\beta R^3/\kappa \approx 0.10\text{--}0.13 \ll 1$. The Rossby stationarity mechanism does not apply; the relevant bridge is Thomson/Onsager.

The binding constraint is *packing*: Neptune’s polar ring radius ($R_{\text{ring}} \approx 5,000 \text{ km}$, estimated from the polar vortex extent) is too small for many cyclones of Jupiter-like size to fit. With the β -drift exclusion factor 1.3 (Gavriel and Kaspi 2021), the effective exclusion radius is $r_{\text{excl}} = 1.3 R_d$, and the prediction depends on the deformation radius:

R_d (km)	$r_{\text{excl}}/R_{\text{ring}}$	N_{pack}	N_{pred}
1,500	0.39	7	≤ 7
2,000 (moderate)	0.52	5	5
2,500 (Jupiter-like)	0.65	4	4

Neptune’s strong internal heat flux ($F_{\text{int}} = 0.43 \text{ W m}^{-2}$, $2.6\times$ absorbed solar) favours vigorous convection, suggesting $R_d \gtrsim 1,000 \text{ km}$. In this regime, the prediction narrows to $N \in \{3, 4, 5\}$ —a triangle, square, or pentagon.

Neptune’s oblateness ($f = 0.017$) is negligible: $\delta\lambda_3 \lesssim 0.01$, far below the Thomson eigenvalue margin.

This prediction is testable by a Neptune polar orbiter (Trident-class mission): imaging the polar region at $\sim 100 \text{ km}$ resolution would resolve individual cyclones and determine N directly.

Falsifiability: a Neptune observation of $N \geq 6$ would violate the packing bound for any $R_d > 1,500 \text{ km}$.

9.3 Uranus: no polygon predicted

Uranus (axial tilt 98°) presents a qualitatively different case. The extreme tilt disrupts the Onsager condensation mechanism by preventing a persistent polar jet: solar forcing alternates between 42-year polar day and 42-year polar night, and the pole receives *more* cumulative insolation than the equator over an orbit. The prediction is $N = 0$ —no polar polygon. This argument is qualitative: there is no sharp tilt threshold in the framework at which the Onsager mechanism switches off. The 98° tilt is invoked as an extreme case where the persistent differential heating needed for a polar jet is absent; a quantitative criterion (e.g. minimum jet persistence time as a function of obliquity) remains to be derived.

Current observations are consistent: [Akins et al. \[2023\]](#) detect a single polar cyclone at Uranus’s north pole ($N = 1$, below the polygon threshold). No ring structure has been observed.

Uranus’s oblateness ($f = 0.023$) is negligible for the same reason as Neptune.

Falsifiability: a Uranus polar polygon would challenge the tilt-disruption hypothesis.

9.4 Why the hexagon is uniquely two-dimensional

The hexagon ($N = 6$) is marginally stable in $d = 2$ ($\lambda_{\min} = +0.5$) but unstable for $d \geq 3$ ($\lambda_{\min} = -0.018$ at $d = 3$):

d	$\Delta = (d-2)/2$	N_{crit}	$\lambda_{\min}(N=6)$	Hexagon
2	0	7	+0.50	Stable
3	1/2	5	−0.02	Unstable
4	1	5	−1.08	Unstable

(The $d \geq 3$ eigenvalues use the csc^d interaction kernel normalised to match the $d = 2$ self-energy at the polygon radius; see Code Availability.)

Saturn’s hexagon requires the logarithmic interaction ($d = 2$, $\Delta = 0$), the Green’s function of the 2D Laplacian. The logarithmic interaction survives in a quasi-2D atmosphere because the weather layer is thin ($H_{\text{eff}} \ll R_{\text{planet}}$) and the barotropic mode dominates. This explains why hexagonal features are observed only on gas giants with thin weather layers and not on terrestrial planets or in 3D laboratory flows.

10 The constraint-intersection principle

The observed polygon wavenumber N on each planet is determined by the intersection of three constraints, all descending from the 2D Laplacian.

Proposition 10.1 (Constraint-intersection selection). *The selected polygon wavenumber is*

$$N_{\text{selected}} = \max \left\{ N : N \leq N_{\text{Thomson}}, \sin(\pi/N) \geq r_{\text{excl}}/R, n \text{ Rossby-stationary} \right\}, \quad (10.1)$$

where inapplicable constraints are dropped. The three constraints are:

1. **Rossby stationarity** (derived; for jet meanders): the QGPV eigenvalue problem $(\nabla^2 + \beta \partial_y)\psi = 0$ selects the stationary wavenumber n^* .

2. **Thomson stability** (derived; for discrete cyclone rings): $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$ gives an upper bound $N \leq N_{\text{max}}(\kappa_0/\kappa)$; the energy monotonicity $H_{N+1} > H_N$ (Proposition 10.2) then selects $N = N_{\text{max}}$.
3. **Geometric packing** (derived skeleton + empirical factor): domain admissibility of $G(\nabla^2)$ on the punctured domain gives $\sin(\pi/N) \geq \varepsilon/R$ where ε is the vortex core radius (derived). The effective exclusion radius $r_{\text{excl}} \approx 1.3\varepsilon$ includes a β -drift buffer zone from nonlinear vortex dynamics [Gavriel and Kaspi, 2021]; the factor 1.3 is observational, not derivable from the linear operator.

Each planetary system activates a different subset of the three constraints:

System	N_{Thomson}	N_{Rossby}	N_{pack}	Binding
Saturn (76°N)	7	6	—	Rossby
Jupiter north	8	—	> 8	Thomson
Jupiter south	8	—	5	Packing

The two Laplacian-derived facets and their observational supplement are: (i) spectral positivity of $G(\nabla^2)$: Havelock eigenvalue $\lambda_m > 0 \rightarrow$ Thomson bound (derived); (ii) domain admissibility of $G_\varepsilon(\nabla^2)$: disc non-overlap $\sin(\pi/N) \geq \varepsilon/R \rightarrow$ packing bound with $r_{\text{excl}} = \varepsilon$ (derived), the enlargement $r_{\text{excl}} \approx 1.3\varepsilon$ at Jupiter south is observational; (iii) Rossby stationarity of $(\nabla^2 + \beta \partial_y)\psi = 0$: stationary mode $n^* \rightarrow$ wave quantization (derived). The finite-core regularisation G_ε does *not* create a blob-stable window for $N = 8$: the corrected constrained Hessian remains negative throughout the admissible domain $\varepsilon/R < \sin(\pi/8)$. The $N = 8$ selection therefore requires the central vortex ($\kappa_0/\kappa \geq \frac{1}{2}$); blob regularisation alone does not provide an alternative stabilisation mechanism.

10.1 Energy monotonicity and maximum- N selection

Proposition 10.2 (Energy monotonicity and maximum- N selection). *Let $\Gamma = N\kappa$ be the total ring circulation, $L = N\kappa R^2$ the angular impulse (so R is fixed when Γ and L are fixed), and $r = \kappa_0/\kappa$ held fixed as N varies (both the ring and central circulations scale proportionally). Denote by r_{core} the vortex core radius—the minimum separation at which the point-vortex model is valid; equivalently, r_{core} is the unit of length in which the logarithm is taken, so that $\ln R > 0$ iff $R > r_{\text{core}}$. Then*

$$H_{N+1} - H_N = \Gamma^2 \left[\frac{2r - 1}{2N(N+1)} \ln R + \frac{1}{2} \left(\frac{\ln N}{N} - \frac{\ln(N+1)}{N+1} \right) \right]. \quad (10.2)$$

For $r \geq \frac{1}{2}$ and $R > r_{\text{core}}$, both terms are non-negative (first: $2r - 1 \geq 0$ and $\ln R > 0$; second: $\ln N/N$ is strictly decreasing for $N > e \approx 2.7$), so $H_{N+1} > H_N$ for all $N \geq 3$. Therefore the N -gon ring energy is strictly monotone in N : the maximum- H configuration within the Thomson-stable set is the ring with the largest N :

$$N_{\text{max}}(\kappa_0/\kappa) \equiv \max \left\{ N' : \kappa_{\text{crit}}(N') \leq \kappa_0/\kappa \right\}.$$

Any dynamics that preferentially populates higher-energy stable configurations—whether Onsager negative-temperature equilibrium ($\beta < 0$, where the Boltzmann weight $e^{-\beta H}$ increases with H), sequential vortex injection, or merger of smaller vortex clusters—selects

$N = N_{\max}$. The stability threshold $\kappa_{\text{crit}}(8) = \frac{1}{2}$ is exactly the value at which $H_{N+1} > H_N$ becomes unconditional (R -independent): at $r = \frac{1}{2}$, term 1 vanishes identically and term 2 remains strictly positive.

Proof. Use the Hamiltonian $H = -\kappa^2 \sum_{j < k} \ln |z_j - z_k| - \kappa_0 \kappa \sum_j \ln |z_j|$ with $\kappa = \Gamma/N$ and $\kappa_0 = r\kappa = r\Gamma/N$.

Step 1: Product formula. For the regular N -gon $z_j = Re^{2\pi i j/N}$, the chord length between vortices j and k is $|z_j - z_k| = 2R \sin(\pi|j - k|/N)$. By $\mathbb{Z}_N$ symmetry the pair sum equals $N/2$ times the sum from one vertex:

$$\sum_{j < k} \ln |z_j - z_k| = \frac{N}{2} \sum_{s=1}^{N-1} \ln(2R \sin(\pi s/N)) = \frac{N}{2} [\ln N + (N-1) \ln R].$$

Here we used $\sum_{s=1}^{N-1} \ln(2 \sin(\pi s/N)) = \ln N$, which follows from evaluating the polynomial identity $(z^N - 1)/(z - 1) = \prod_{k=1}^{N-1} (z - e^{2\pi i k/N})$ at $z = 1$: the product of moduli is $\prod_{s=1}^{N-1} |1 - e^{2\pi i s/N}| = \prod_{s=1}^{N-1} 2 \sin(\pi s/N) = N$.

Step 2: Central vortex term. Since $|z_j| = R$ for all j : $\sum_j \ln |z_j| = N \ln R$.

Step 3: Closed form for H_N . Substituting Steps 1–2 with $\kappa = \Gamma/N$, $\kappa_0 = r\Gamma/N$:

$$H_N = -\frac{\Gamma^2}{2N} \ln N - \Gamma^2 \left(\frac{N-1}{2N} + \frac{r}{N} \right) \ln R. \quad (10.3)$$

Step 4: Difference $H_{N+1} - H_N$. Writing H_{N+1} from (10.3) with $N \rightarrow N+1$ and subtracting:

$$H_{N+1} - H_N = -\frac{\Gamma^2}{2} \left[\frac{\ln(N+1)}{N+1} - \frac{\ln N}{N} \right] - \Gamma^2 \ln R \left[\frac{N/2 + r}{N+1} - \frac{(N-1)/2 + r}{N} \right].$$

The coefficient of $\ln R$ simplifies by partial fractions:

$$\frac{N/2 + r}{N+1} - \frac{(N-1)/2 + r}{N} = \frac{N^2 - (N-1)(N+1)}{2N(N+1)} - \frac{r}{N(N+1)} = \frac{1 - 2r}{2N(N+1)},$$

where we used $N^2 - (N-1)(N+1) = 1$. Negating and rearranging gives (10.2).

Step 5: Sign of each term. *Term 2:* $f(x) = \ln x/x$ satisfies $f'(x) = (1 - \ln x)/x^2 < 0$ for $x > e$, so $\ln N/N > \ln(N+1)/(N+1)$ for all $N \geq 3$.

Term 1: $2r - 1 \geq 0$ for $r \geq \frac{1}{2}$; and $\ln R > 0$ for $R > r_{\text{core}}$. Hence both terms are non-negative, and term 2 is strictly positive, giving $H_{N+1} > H_N$. $\square$

Remark 10.1 (Physical status of the fixed- r assumption). The assumption $r = \kappa_0/\kappa$ fixed as N varies means that as more cyclones appear in the ring (N increases), each individual ring vortex becomes weaker ($\kappa = \Gamma/N$) and the central vortex weakens proportionally ($\kappa_0 = r\Gamma/N$). The assumption is justified when all vortices are drawn from the same turbulent condensation process, so their strengths are set by a common large-scale energy budget. In the Onsager picture, the negative-temperature state is reached by the inverse cascade from the same vorticity reservoir, giving a fixed partition of total circulation between center and ring. If instead κ_0 is held fixed while $\kappa = \Gamma/N$ varies, then $r = N\kappa_0/\Gamma$ grows with N and the monotonicity strengthens further, so maximum- N selection remains qualitatively intact. The marginal case is $\kappa_0 = 0$: then $r = 0 < \frac{1}{2}$ and Term 1 in (10.2) is negative, so $H_{N+1} > H_N$ requires R large enough that Term 2 dominates. The $r \geq \frac{1}{2}$ condition is therefore specific to the regime with a sufficiently strong central vortex.

The $N = 8$ selection is robust across both limiting assumptions (fixed r and fixed κ_0) but relies on the Onsager framework providing the link between N , Γ , and κ_0 —a physical hypothesis, not a mathematical consequence.

For Jupiter north, $\kappa_0/\kappa \approx 1$ with roughly 30% observational uncertainty (Juno; [Adriani et al. 2018](#)), giving a conservative range $\kappa_0/\kappa \in [0.7, 1.3]$. Since $\kappa_{\text{crit}}(8) = \frac{1}{2}$ exactly, this entire range satisfies the stability condition for $N = 8$. However, $\kappa_{\text{crit}}(9) = 1$, which lies *inside* the uncertainty interval: if $\kappa_0/\kappa \geq 1$, then $N = 9$ is also Thomson-stable and the Onsager argument would select $N_{\text{max}} = 9$. The conclusion $N_{\text{max}} = 8$ relies on $\kappa_0/\kappa < 1$, which is consistent with the central estimate but not guaranteed by current observational precision. The observed $N = 8$ is thus evidence that $\kappa_0/\kappa \lesssim 1$, providing a theory-derived constraint on the central vortex strength testable with improved Juno wind measurements.

10.2 Jupiter’s north–south asymmetry

For Jupiter’s south pole ($N = 5$): $\kappa_{\text{crit}}(N) = 0$ for all $N \leq 7$, so the Thomson bound alone allows up to $N = 7$ regardless of the central vortex. The observed $N = 5$ is selected by the geometric packing constraint (Proposition 10.1, item 3). South-pole cyclones are individually $\approx 40\%$ larger in linear size than north-pole cyclones ($r_{\text{core}} \approx 3,500$ km vs. 2,500 km; [Adriani et al. 2018](#)), and each cyclone carries a β -drift exclusion zone extending $\approx 30\%$ beyond the core radius [[Gavriel and Kaspi, 2021](#)], giving $r_{\text{excl}} \approx 1.3 r_{\text{core}} \approx 4,550$ km. At the south-pole ring radius $R \approx 8,000$ km ($\sim 7^\circ$ from the pole):

$$\sin(\pi/5) = 0.588 \geq \frac{r_{\text{excl}}}{R} = \frac{4,550}{8,000} = 0.569, \quad \sin(\pi/6) = 0.500 < 0.569.$$

Thus $N_{\text{pack}} = 5$, which is tighter than the Thomson bound $N_{\text{Thomson}} = 8$: packing, not stability, is the binding constraint at Jupiter’s south pole. The selection is tight: $(\sin(\pi/5) - r_{\text{excl}}/R)/(\sin(\pi/5)) = 3.3\%$.

The entire $N = 8$ vs. $N = 5$ asymmetry reduces to a single dimensionless ratio:

$$\chi = \frac{R_d^{(\text{south})}}{R_d^{(\text{north})}} \approx \frac{r_{\text{cyclone}}^{(\text{south})}}{r_{\text{cyclone}}^{(\text{north})}} \approx \frac{3,500}{2,500} = 1.40.$$

The constraint intersection requires $\chi \gtrsim 1.23$ for the south-pole packing bound to tighten from $N = 6$ to $N = 5$. The observed $\chi = 1.40$ matches to 8%. Deriving χ from atmospheric physics (insolation asymmetry, moist convection) remains open: radiative scaling gives $\chi \sim 1.07$ (insufficient), suggesting that deep convective dynamics set the pole-to-pole deformation-radius contrast.

10.3 Saturn’s structural stability

The Rossby stationarity argument selects Saturn’s wavenumber $n = 6$ (§8). A natural question: can smooth perturbations of the jet destroy the stationary wave, converting the hexagonal meander into an evanescent mode? The answer is no, by a WKB branch-disconnection argument.

The Rayleigh–Kuo equation for stationary ($c = 0$) mode n on a zonal jet $U(y)$ reads $\hat{\psi}'' + V(\rho)\hat{\psi} = 0$, where $V = (\beta - U'')/U - n^2/R^2$ is the effective potential in log-polar

coordinates. At the jet center ($y = 0$) of a Gaussian jet:

$$V(0) = \frac{\beta + U_{\max}/\sigma^2}{U_{\max}} - \frac{n^2}{R^2} = \frac{\beta_{\text{eff}}}{U_{\max}} - \frac{n^2}{R^2}.$$

For Saturn ($U_{\max} = 120$ m/s, $\sigma = 2.5 \times 10^6$ m, $\beta = 1.46 \times 10^{-12}$ m⁻¹s⁻¹, $n = 6$, $R = R_S \cos 76^\circ = 1.31 \times 10^7$ m): $V(0) = 1.72 \times 10^{-13} - 2.10 \times 10^{-13} = -3.8 \times 10^{-14}$ m⁻² (negative). The sign of $V(0)$ depends on σ_{jet} : $V(0) < 0$ for $\sigma_{\text{jet}} \gtrsim 2.3 \times 10^6$ m, with a $\sim 9\%$ margin above the sign change $((\sigma - \sigma_{\text{crit}})/\sigma_{\text{crit}})$ at the adopted $\sigma_{\text{jet}} = 2.5 \times 10^6$ m (Cassini jet profile measurements constrain $\sigma_{\text{jet}} \in [2.3, 3.0] \times 10^6$ m).

Since $V < 0$, the WKB exponent $s^2 = V < 0$ forces oscillatory solutions with $s = i\alpha_\rho$ ($\alpha_\rho \neq 0$). Writing $s = \sigma + i\alpha_\rho$: $\text{Im}(s^2) = 2\sigma\alpha_\rho = 0$. When $\alpha_\rho \neq 0$ (oscillatory region): $\sigma = 0$. The elliptic ($\sigma = 0$) and hyperbolic ($\alpha_\rho = 0$) branches are disconnected: smooth jet perturbations cannot move the solution from the oscillatory branch to the evanescent branch at leading WKB order. No continuous deformation of the jet profile can remove the $n = 6$ stationary wave without first passing through $V(0) = 0$ (a finite perturbation). The hexagon is therefore structurally stable at leading WKB order.

10.4 Quality metrics and cross-planetary comparison

The two planetary polygon types require different quality metrics. For *discrete vortex rings* (Jupiter), the natural measure is the geometric deviation from a perfect N -gon:

$$\sigma_{\text{geom}} = \frac{1}{N} \sum_{k=1}^N \left| \ln |z_k/z_k^{\text{ideal}}| \right|, \quad (10.4)$$

where z_k^{ideal} are the best-fit regular N -gon vertices. For *continuous jet meanders* (Saturn), the appropriate measure is the fractional radial displacement of the jet center:

$$\varepsilon_{\text{obs}} = A_n/\bar{R}, \quad (10.5)$$

where A_n is the half-amplitude of the n -th Fourier mode and $\bar{R}$ is the mean ring radius. For Saturn’s hexagon ($n = 6$, $\bar{R} \approx 13,200$ km), $A_6 \approx 1,900$ km and $\varepsilon_{\text{obs}} \approx 0.14$ [Godfrey, 1988, Antuñano et al., 2015]. These two metrics are not interchangeable: applying σ_{geom} to Saturn yields 0.50, an order of magnitude larger than Jupiter’s 0.05–0.06, but this conflates meander amplitude with positional scatter.

Table 1: Planetary polygon configurations. κ_0/κ is estimated from Juno imagery [Adriani et al., 2018] using cyclone area as a proxy ($\sim 30\%$ systematic uncertainty). “—” indicates the quantity is not applicable.

System	N	Quality metric	κ_0/κ	Selection mechanism
Saturn hexagon	6	$\varepsilon_{\text{obs}} = 0.14$	N/A (jet)	Rossby stationarity
Jupiter north	8	$\sigma_{\text{geom}} = 0.06$	~ 1	Thomson + Onsager
Jupiter south	5	$\sigma_{\text{geom}} = 0.05$	—	Packing (Prop. 10.1)

Independent numerical support comes from Siegelman et al. [2022], who produce $N = 8$ vortex crystals from 2D turbulence simulations on polar caps, with a central polar cyclone emerging naturally alongside the ring. These simulations include continuous small-scale forcing and large-scale dissipation, yet the polygon crystal is the persistent stationary state—numerical evidence that the constrained energy minimum acts as a dynamical attractor under turbulent forcing within the 2D barotropic framework.

11 Circulation disorder and stability domain robustness

11.1 Resilience to circulation disorder

In both planetary regimes the vortex circulations κ_k are not perfectly equal; convection injects or removes circulation from individual vortices continuously. We model this as $\kappa_k = \kappa(1 + \eta_k)$ with i.i.d. fractional disorders $\eta_k \sim \mathcal{N}(0, \sigma_\eta^2)$.

We work on $\mathbf{H}^2$ rather than the flat plane because the hyperbolic setting simultaneously tests two effects: circulation disorder and curvature stabilization. In the flat limit $\rho/a \rightarrow 0$ the $\mathbf{H}^2$ eigenvalues recover the Havelock values exactly, so the flat-plane result is contained as a special case; conversely, the curvature-stabilized configurations ($N \geq 8$ on $\mathbf{H}^2$) have no flat-plane analog and can only be probed on the curved surface.

On $\mathbf{H}^2$, the disordered Hamiltonian is

$$H_\kappa = - \sum_{j < k} \kappa_j \kappa_k \ln |z_j - z_k| + \frac{1}{2} \sum_j \kappa_j (K_{\text{tot}} - \kappa_j) \ln(a^2 - |z_j|^2), \quad (11.1)$$

where $K_{\text{tot}} = \sum_k \kappa_k$. At $\eta_k \equiv 0$ this reduces exactly to the uniform Hamiltonian H_{hyp} .

Holding the ring positions at the uniform equilibrium (exact to $O(\eta)$; the $O(\eta)$ shift of equilibrium produces only an $O(\eta^2)$ eigenvalue correction), we compute the minimum Fourier-mode eigenvalue $\lambda_{\min}(\eta)$ via finite differences. Monte Carlo over $n = 1000$ realizations estimates $P(\lambda_{\min} < 0)$ as a function of (σ_η, N, ρ) .

Physical range of σ_η . Juno imagery [Adriani et al., 2018, Gavriel and Kaspi, 2021] shows Jupiter’s eight north-polar cyclones have peak wind speeds ranging from ~ 60 to $\sim 100 \text{ m s}^{-1}$ and diameters from $\sim 4,000$ to $\sim 7,000 \text{ km}$. Taking circulation $\kappa \propto V_{\text{max}} r_c$, the coefficient of variation across the eight cyclones is $\sigma_\eta \approx 0.15\text{--}0.25$. Saturn’s hexagonal jet is more uniform (Cassini wind profiles give $\sigma_\eta \lesssim 0.05$ for the azimuthal circulation variation). The physically relevant range is therefore $\sigma_\eta \in [0.05, 0.25]$.

Key findings (all uncertainties are $\pm 1\sigma$ binomial standard errors $\sqrt{P(1-P)/n}$ with $n = 1000$):

- **Stable configurations are robust.** For $N = 6$ at $\rho = 0.5a$, the instability probability remains below 10% ($\pm 0.9\%$) for $\sigma_\eta \leq 0.2$, covering the full physically relevant range.
- **Curvature-stabilized rings ($N \geq 8$) are similarly robust.** At $N = 8$, $\rho = 2a$ (well inside the stable region on $\mathbf{H}^2$), $P < 0.5$ ($\pm 1.6\%$) for $\sigma_\eta \leq 0.1$. At $\sigma_\eta = 0.2$ (Jupiter’s estimated value), P remains below 15%.
- **The $N = 7$ marginal ring is sensitive.** At $N = 7$, $\rho = 0.05a$ (flat-limit regime), $P(\lambda_{\min} < 0)$ increases monotonically with σ_η : $P \approx 0$ ($\pm 0.3\%$) at $\sigma_\eta = 0.1$ and $P > 0$ for $\sigma_\eta \geq 0.5$. The marginal mode ($\lambda_0 = 0$ at $K = 0$) is the first to be destabilized by disorder, as expected. At Jupiter’s estimated $\sigma_\eta \approx 0.2$, a hypothetical $N = 7$ ring would have $P \approx 0$ —the marginal sensitivity manifests only at unphysically large disorder.

Thus the N_{crit} boundary is a *stability domain boundary*, not a *knife-edge*: configurations comfortably inside the stable region remain stable under the circulation fluctuations observed at Jupiter ($\sigma_\eta \approx 0.2$) and Saturn ($\sigma_\eta \lesssim 0.05$).

See Code Availability.

11.2 The regime criterion

The dimensionless number $\beta R^3/\kappa$ is the ratio of two timescales: the Rossby wave period $T_R \sim 1/(\beta R)$ and the vortex turnover time $T_v \sim R^2/\kappa$. When $\beta R^3/\kappa = T_v/T_R \ll 1$, the vortex interactions complete many orbits before the Rossby wave propagates appreciably, so the dynamics is vortex-dominated. When $\beta R^3/\kappa \gg 1$, the Rossby wave dynamics operates faster than the vortex rearrangement, so the wave mechanism dominates:

Regime	$\beta R^3/\kappa$	Bridge
Vortex-dominated	$\ll 1$	Blob convergence (Proposition 7.1)
Wave-dominated	$\gg 1$	Rossby stationarity (equation (7.5))

Jupiter's north pole. Throughout this regime calculation κ denotes the *Rankine-core* circulation, which integrates only within the solid-body rotation core and sets the characteristic vortex timescale $T_v \sim R^2/\kappa$. The full-cyclone circulation κ_{full} (which includes the outer irrotational skirt) is larger and is the appropriate quantity for computing orbital periods and eigenfrequencies; it is used there instead. The two estimates are numerically distinguished below.

At 84°N , the planetary vorticity gradient is $\beta_J = 2\Omega_J \cos(84^\circ)/R_J \approx 5.1 \times 10^{-13} \text{ m}^{-1}\text{s}^{-1}$ and the cyclone-ring radius is $R \approx 7,470 \text{ km}$. Each Juno-observed circumpolar cyclone has Rankine core radius $r_c \approx 700 \text{ km}$ and peak wind $V_{\text{max}} \approx 80 \text{ m s}^{-1}$ [Adriani et al., 2018, Gavriel and Kaspi, 2021], giving $\kappa_{\text{Rank}} = 2\pi r_c V_{\text{max}} \approx 3.5 \times 10^8 \text{ m}^2 \text{ s}^{-1}$ (full-cyclone estimate: $r_c \approx 3,200 \text{ km}$, $\kappa_{\text{full}} \approx 1.6 \times 10^9 \text{ m}^2 \text{ s}^{-1}$). Hence

$$\frac{\beta R^3}{\kappa} = \frac{5.1 \times 10^{-13} (7.47 \times 10^6)^3}{3.5 \times 10^8} \approx 0.61.$$

This places Jupiter in the vortex-dominated regime, but only moderately: 0.61 is $O(1)$, not $\ll 1$. Rossby-wave effects are not negligible at this value, and a decrease in the effective κ (e.g. using the Rankine-core rather than full-cyclone estimate) would push the ratio above unity. The qualitative conclusion—discrete cyclones, Thomson stability applies, blob bridge proves convergence—is robust to this uncertainty because the vortex-dominated bridge requires only that the cyclones be identifiable as N distinct vortices (which the Juno imagery unambiguously confirms), not that $\beta R^3/\kappa$ be parametrically small. The ratio enters quantitatively only through the Rossby-wave correction to the eigenvalues, which is $O(\beta R^3/\kappa)$ and does not change the sign of λ_{min} for $N = 8$ (where $|\lambda_{\text{min}}| = 1 > 0.61$). Using the full-cyclone circulation $\kappa_{\text{full}} \approx 1.6 \times 10^9 \text{ m}^2 \text{ s}^{-1}$ gives $\beta R^3/\kappa \approx 0.13$, firmly in the vortex-dominated regime.

Saturn's hexagon. At 76°N , $\beta_S = 2\Omega_S \cos(76^\circ)/R_S \approx 1.3 \times 10^{-12} \text{ m}^{-1}\text{s}^{-1}$ and $R \approx 14,600 \text{ km}$. Saturn's polar jet is well-described by a Gaussian profile $U = U_{\text{max}} \exp(-y^2/2\sigma^2)$; its effective circulation is $\kappa = U_{\text{max}} \sqrt{2\pi} \sigma$. With $U_{\text{max}} \approx 100 \text{ m s}^{-1}$ and $\sigma \approx 2,100 \text{ km}$ (Cassini VIMS wind profiles), $\kappa \approx 5.4 \times 10^8 \text{ m}^2 \text{ s}^{-1}$, giving

$$\frac{\beta R^3}{\kappa} = \frac{1.3 \times 10^{-12} (1.46 \times 10^7)^3}{5.4 \times 10^8} \approx 7.6.$$

Continuous jet meander, Rossby stationarity selects $n = 6$, periodicity provides quantization.

Both mechanisms require the 2D logarithmic Green’s function as the common mathematical kernel.

12 Predictions for quantum systems

12.1 BEC vortex clusters

In a quasi-2D Bose–Einstein condensate in a circular trap, quantized vortices interact logarithmically at scales above the healing length ξ . The circulation is quantized: $\kappa = q \cdot h/m$ with integer winding number q . The κ_0/κ ratio of Proposition 4.1 becomes the winding number ratio q_0/q , yielding sharp integer-valued predictions:

Configuration	κ_{crit}	Prediction
$N \leq 7$ ring	0	Stable without central vortex
$N = 8$ ring	$1/2$	Unstable alone; stable with $q_0 \geq 1$
$N = 9$ ring	1	Marginally stable with $q_0 = 1$; stable with $q_0 \geq 2$
$N = 10$ ring	$7/4$	Unstable with $q_0 = 1$; stable with $q_0 \geq 2$

The $N = 8$ prediction is the sharpest: a ring of 8 unit-winding vortices should be unstable alone and stable upon adding a single central vortex ($q_0 = 1 > \kappa_{\text{crit}} = 1/2$).

The $N = 7$ case offers a distinct signature. The quartic restoring force ($\alpha_0 = 45/14$, Section 5) produces an *amplitude-dependent* oscillation frequency for the $m = 3$ mode. In the normal form $\mathcal{H} = (\alpha_0/4)\varepsilon^4$, the nonlinear oscillation frequency is $\omega_{m=3} = \alpha_0 \varepsilon^2 \kappa / (2\pi R^2)$, where ε is the dimensionless perturbation amplitude (radial displacement / ring radius) and $\kappa = h/m_{\text{atom}}$ is the single-quantum circulation. For ^{87}Rb in a circular trap of radius $R = 50 \mu\text{m}$: $\kappa \approx 1.2 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$, $\kappa / (2\pi R^2) \approx 7.6 \text{ s}^{-1}$, so $\omega_{m=3} \approx 24 \varepsilon^2 \text{ s}^{-1}$. At a 5% radial displacement ($\varepsilon = 0.05$): $\omega_{m=3} \approx 0.06 \text{ s}^{-1}$ (period $\sim 100 \text{ s}$), resolvable with standard BEC vortex imaging at $\sim 10 \text{ Hz}$ frame rates. The key signature distinguishing the $N = 7$ quartic confinement from the $N \leq 6$ quadratic regime is that the $m = 3$ oscillation frequency *vanishes* as $\varepsilon \rightarrow 0$ (rather than approaching a finite value), producing an amplitude-dependent period that is directly measurable.

The healing length ξ provides a natural blob width $\varepsilon_{\text{blob}} = \xi/R$. For typical BEC parameters ($\xi \sim 0.3 \mu\text{m}$, $R \sim 50 \mu\text{m}$): $\varepsilon_{\text{blob}} \approx 0.006$, well within the point vortex regime. The blob correction (Proposition 7.1) is negligible: $\lambda(0.006) \approx \lambda_0 + O(10^{-4})$.

Finite-temperature effects. Current BEC vortex experiments operate at finite temperature ($T/T_c \sim 0.3\text{--}0.7$), where a thermal cloud coexists with the condensate and exerts a dissipative drag on vortex motion [Fetter, 2009]. This drag damps the oscillation modes on a timescale $\tau_{\text{drag}} \sim R^2 / (D_v T/T_c)$ where D_v is the vortex diffusion coefficient. For the $N = 7$ quartic mode to be observable, the oscillation period ($\sim 100 \text{ s}$ at $\varepsilon = 0.05$) must be shorter than τ_{drag} ; at $T/T_c = 0.5$ this requires $D_v \lesssim 10^{-9} \text{ m}^2 \text{ s}^{-1}$, which is achievable in large-radius traps ($R \geq 50 \mu\text{m}$) at low temperature. The $N = 8$ stability prediction (unstable alone, stable with $q_0 = 1$) is less sensitive to dissipation because it concerns exponential growth vs. decay, not oscillation frequency; finite temperature adds a damping channel but does not change the sign of the eigenvalue.

The hard-wall disk: an exact theorem. In a hard-wall circular trap of radius R , each vortex at z has an image at $R^2/\bar{z}$. The effective interaction is *exactly* the Green's function on the Poincaré disk of radius R (the conformal equivalence is the identity map).

Proposition 12.1 (Hard-wall BEC = Poincaré disk). *In a hard-wall circular BEC of radius R , the stability eigenvalue of an N -gon vortex ring at radius r is $\lambda_m = C_1(\xi) - m(N-m)/2$ with $\xi = r^2/R^2$ and $C_1(\xi) = (N-1)(1+\xi^2)/(1-\xi)^2$ —the same formula as for $\mathbf{H}^2$ (Paper I, equation 5.7).*

Proof. The image vortex construction gives the Green's function $G(z, w) = -(1/2\pi) \log |z-w| + (1/2\pi) \log |1-z\bar{w}/R^2|$, which is exactly the Green's function on the Poincaré disk of radius R . The Havelock decomposition applies without approximation. $\square$

The palindromic thresholds $\xi^*(N)$ are therefore *exact* predictions for hard-wall BEC experiments:

N	$\xi^* = r^2/R^2$	r/R	Prediction
≤ 7	0	0	Stable at any radius
8	$8-3\sqrt{7} \approx 0.063$	0.25	Stable for $r > 0.25 R$
11	$(1+\sqrt{2})^{-2} \approx 0.17$	0.41	Stable for $r > 0.41 R$
23	$\varphi^{-2} \approx 0.38$	0.62	Stable for $r > 0.62 R$

The **key testable prediction**: an 8-vortex ring at $r/R > 0.25$ in a hard-wall disk BEC should be stable; at $r/R < 0.25$ it should be unstable to the $m = 4$ mode. The threshold $r/R = \sqrt{8-3\sqrt{7}} \approx 0.25$ is an algebraic number in $\mathbb{Q}(\sqrt{7})$ —the same field that governs the $7 \rightarrow 8$ transition in planetary vortex dynamics.

Flat-bottomed BEC traps, in which the point vortex model has been experimentally validated [Gauthier et al., 2019], are the natural setting for testing these predictions.

The harmonic trap: an approximation. For a harmonic trap $V(r) = \frac{1}{2}m\omega^2 r^2$ with Thomas–Fermi radius R_{TF} : the identification $\xi = r^2/R_{\text{TF}}^2$ holds to leading order, but the soft boundary introduces corrections at $O(\xi^2)$ relative to the hard-wall case (the image vortex positions shift by terms proportional to the trap anharmonicity). The prediction table above applies for $\xi \ll 1$ ($r \ll R_{\text{TF}}$), where the hard-wall and harmonic cases coincide. For vortex rings near the condensate edge ($\xi \rightarrow 1$), the harmonic trap softens the effective C_1 , and the point vortex model itself requires corrections from the inhomogeneous density profile.

Healing-length and quantum corrections. The BEC healing length ξ_{heal} sets the blob scale $\varepsilon = \sqrt{2} \xi_{\text{heal}}/d$ where $d = 2r \sin(\pi/N)$ is the intervortex spacing. For typical parameters ($\xi_{\text{heal}} \approx 0.3 \mu\text{m}$, $R \approx 50 \mu\text{m}$): $\varepsilon \approx 0.02$, giving $\delta\lambda \sim \varepsilon^2 \sim 0.04\%$ (Proposition 7.1).

The *quantum* correction (from the Bogoliubov–de Gennes excitation spectrum) is even smaller. The BdG anomalous coupling between vortex cores separated by d is $O(\exp(-d/\xi_{\text{heal}}))$; for $N = 8$ at the threshold ($d \approx 9.6 \mu\text{m}$): $\exp(-d/\xi_{\text{heal}}) \approx 10^{-14}$. The effective mass renormalization shifts the precession frequency but not the energy Hessian (the stability condition $\lambda_m > 0$ is a property of the potential landscape, not the kinetic term). The palindromic threshold $\xi^*(8) = 8-3\sqrt{7}$ is therefore an *exact* feature of the quantum spectrum, preserved to exponential accuracy by the topological invariance of the K_0 class (Paper I, Theorem 6.11).

Hyperbolic lattice realisations. Interacting bosons on a $\{p, q\}$ hyperbolic lattice (Bose–Hubbard model, hopping t , on-site U) reduce to the point vortex Hamiltonian in the

Thomas–Fermi regime ($U \gg t$): the mean-field condensate $\psi = \sqrt{\rho} e^{i\theta}$ has vortices where $\psi = 0$, and θ satisfies $\Delta_{\text{lattice}}\theta = 2\pi \sum_k \delta(z - z_k)$, giving the lattice Green’s function interaction $H = -\sum_{j < k} G_{\text{lattice}}(z_j, z_k)$. In the continuum limit (lattice spacing $a \ll$ intervortex spacing d): $G_{\text{lattice}} \rightarrow G_{\mathbf{H}^2}$ and the Havelock thresholds apply. The continuum limit requires $\gtrsim 1000$ lattice sites—beyond current circuit-QED (~ 100 sites) but within reach of topoelectric circuits ($\sim 10^4$ sites). On the Bolza tiling ($\{8, 8\}$, genus 2, $|\text{Aut}| = 48$): the prediction is $N_{\text{crit}} = 12$ at the Aut-fixed centre, with $\delta C_1 \approx 0.011$ (Paper I, §7).

12.2 Mesoscopic superconductors

Remark 12.1 (Mesoscopic superconducting disks). *Abrikosov vortices in a type-II superconducting disk of radius R obey the same boundary condition as BEC vortices in a hard-wall trap (supercurrent vanishes at the edge $\Leftrightarrow$ Kelvin images at $R^2/\bar{z}$). Proposition 12.1 therefore applies directly: $\lambda_m = C_1(\xi) - m(N-m)/2$ with $\xi = r_0^2/R^2$. The observed experimental sequence in mesoscopic Nb disks (Grigorieva et al. 2006): single polygon for $N \leq 6$, marginal polygon or shell (1+6) at $N = 7$, and shells (1+7, 2+7, ...) for $N \geq 8$, is consistent with $N_{\text{crit}} = 7$ at the disk centre ($\xi = 0$) and central-vortex stabilisation (Section 4) raising N_{crit} to 8 for the outer ring.*

12.3 Quantum Hall systems

The following two remarks apply the stability framework to fractional quantum Hall states. They are independent of the gravitational interpretation (Papers III–IV) and may be read as standalone applications of the Havelock decomposition.

Remark 12.2 (Laughlin quasihole polygons). *The effective Hamiltonian of the Laughlin state at filling $\nu = 1/m$ is $H_{\text{eff}} = -m \sum_{j < k} \log |z_j - z_k| + (4\ell_B^2)^{-1} \sum |z_k|^2$: the logarithmic energy with harmonic confining. Quasiholes (created by factors $\prod(z_j - \eta_a)$) interact via $H_{\text{qh}} = -(1/m) \sum_{a < b} \log |\eta_a - \eta_b| + \text{confining}$, where the $1/m$ comes from the fractional charge $e^* = e/m$. For P quasiholes in a ring at radius r inside a Laughlin droplet of N electrons (radius $R = \ell_B \sqrt{2N}$): the Havelock eigenvalue is $\lambda_p = (1/m) [C_1(\xi) - p(P-p)/2]$ with $\xi = r^2/R^2$. The factor $1/m$ is overall: the stability threshold P_{crit} is independent of the filling fraction m . The palindromic thresholds of Paper I, §5 apply directly: $P \leq 7$ is stable at any r ; $P = 8$ requires $r/R > 0.25$; etc. The Berry phase (statistical angle $\theta = \pi/m$, giving anyonic braiding) adds an imaginary term to H_{eff} that affects the dynamics but not the energy Hessian—the stability eigenvalues are real and independent of the statistical angle. The conditions: $N \gtrsim 55$ electrons (for $P = 8$ at $r/R = 0.25$), and $d = 2r \sin(\pi/P) \gg \ell_B$ (well-separated quasiholes).*

Remark 12.3 (Non-abelian quasiholes and topological quantum computation). *The Moore–Read state ($\nu = 5/2$) has quasiholes carrying Majorana zero modes with non-abelian braiding. The effective interaction between two quasiholes at separation d is $E(d) = E_{\text{log}}(d) + \delta E_{\text{fusion}}(d)$ where $\delta E_{\text{fusion}} \sim C \exp(-d/\xi_E)$ with $\xi_E \approx 2.3 \ell_B$ (Baraban et al. 2009). The exponential decay gives effective exponent $\alpha = \infty \gg 2$ in the RG classification (Remark 2.1): the fusion correction is deeply irrelevant. By the blob bridge (Proposition 7.1): $P_{\text{crit}} = 7$ is unchanged.*

The same argument applies to all gapped quantum Hall states (Read–Rezayi, hierarchy states): the fusion correction is always exponential in d (from the bulk energy gap), hence

always RG-irrelevant. $P_{\text{crit}} = 7$ is therefore universal across all gapped quantum Hall states, independent of filling fraction, braiding statistics, and fusion channel.

For topological quantum computation: standard braiding protocols use $P \leq 6$ quasi-holes (4 for Moore–Read, 6 for two-qubit Fibonacci), safely below P_{crit} . The threshold is not a practical constraint on current proposals but is a fundamental limit on the complexity of quasihole polygon configurations in any quantum Hall platform.

13 Gravitational polygon stability in 2+1 dimensions

The 2+1D gravity literature studies the single conical deficit [Deser et al., 1984], two-body scattering, BTZ black holes, polygon tessellations of spatial slices ('t Hooft 1993; note that “polygon” there refers to the tiles of the tessellation, not to a configuration of point masses), and one-loop partition functions [Giombi et al., 2008]. What appears not to have been asked is: *for how many point masses can one form a stable regular polygon on a spatial slice, and what algebraic number theory governs the answer?* This section derives the answer from the vortex stability framework of Paper I.

In 2+1 dimensional gravity with negative cosmological constant ($\Lambda = -1/\ell^2$), space-time is locally AdS_3 and has no propagating degrees of freedom. A point mass m creates a conical deficit of angle $\alpha = 8\pi Gm$ [Deser et al., 1984]. A constant-time spatial slice is a quotient $S = \Gamma \backslash \mathbf{H}^2$, and the *linearised* gravitational potential of a point mass on S satisfies $\nabla^2 \Phi = 8\pi G m \delta(z - z_0)$ —the same Laplace equation as for the stream function of a point vortex.

Proposition 13.1 (Gravitational polygon stability). *In the linearised regime ($8\pi Gm_i \ll 2\pi$, small deficit angles), N equal point masses of mass m on a spatial slice $S = \Gamma \backslash \mathbf{H}^2$ interact via*

$$E_{\text{grav}} = -(8\pi G)^2 \sum_{i < j} m^2 G_S(z_i, z_j), \quad (13.1)$$

where G_S is the Green’s function of the Laplacian on S (the same G_S appearing in the vortex energy with $\kappa \leftrightarrow 8\pi Gm$). The regular N -gon is a critical point of E_{grav} on the angular-impulse surface, and the Hessian eigenvalues are

$$\lambda_m^{\text{grav}} = C_1(S, \xi) - \frac{m(N-m)}{2}, \quad m = 1, \dots, N-1, \quad (13.2)$$

with the same C_1 and $m(N-m)/2$ as for vortices. The N -gon is a stable equilibrium ($\lambda_m^{\text{grav}} > 0$ for all m) if and only if $N \leq N_{\text{crit}}(S, \xi)$.

Proof. The linearised gravitational potential satisfies the same Poisson equation as the vortex stream function, with $\kappa = 8\pi Gm$. The interaction energy (13.1) is therefore $E_{\text{grav}} = (8\pi Gm/\kappa)^2 \cdot H_{\text{vortex}}$, a positive multiple of the vortex Hamiltonian. At the N -gon, the first and second variations of E_{grav} are proportional to those of H_{vortex} , so the critical points, the Hessian eigenvalues, and the stability criterion ($\lambda_m > 0$ for all m) are identical. The Havelock decomposition (Paper I, §4) gives (13.2).

Note on dynamics. The vortex equations of motion are *first order* (the Hamiltonian is the stream function), while the gravitational equations are *second order* (kinetic + potential energy). The stability criterion—positive-definite Hessian of the potential—is the same for both: a positive eigenvalue $\lambda_m > 0$ gives stable oscillation in the gravitational case and stable circulation in the vortex case. A negative eigenvalue gives exponential growth in both. The criterion $N \leq N_{\text{crit}}$ is therefore identical. $\square$

Validity and the developing map. For N equal masses in a ring, each creates a deficit angle $\delta = 8\pi Gm$. The total deficit is $N\delta < 4\pi$ (Gauss–Bonnet for S^2), giving $Gm < 1/(2N)$. We now show that the backreaction of the deficit angles on the flat-plane stability is *exactly zero*.

The **developing map** of the flat cone metric with $\mathbb{Z}_N$ symmetry is

$$f(z) = z \cdot {}_2F_1\left(\beta, \frac{1}{N}; 1 + \frac{1}{N}; z^N\right), \quad \beta = 4Gm, \quad (13.3)$$

with derivative $f'(z) = (1 - z^N)^{-\beta}$. This is a conformal isometry from the cone surface to the Euclidean plane: the metric is $|f'|^2 |dz|^2$, flat away from the cone tips, with cone angle $2\pi(1-\beta)$ at each N -th root of unity.

$\mathbb{Z}_N$ symmetry. $f(\omega z) = \omega f(z)$ for $\omega = e^{2\pi i/N}$: the image of the N -gon of cone points is a regular N -gon in the developed plane, with radius

$$R_f = \frac{\Gamma(1 + 1/N) \Gamma(1 - \beta)}{\Gamma(1 + 1/N - \beta)} \quad (13.4)$$

(Gauss’s summation formula for ${}_2F_1$ at $z = 1$). At $\beta = 0$: $R_f = 1$. At the Gauss–Bonnet endpoint $\beta = 1/N$: $R_f = (\pi/N)/\sin(\pi/N)$ (by the Γ -reflection formula). The image radius *increases* with β : the cone metric inflates distances near the tips.

Proposition 13.2 (Conformal invariance of flat-plane stability). *On a flat spatial slice ($\Lambda = 0$) with N equal conical singularities of deficit $\delta = 8\pi Gm$ in a regular ring, the Havelock eigenvalues are*

$$\lambda_m = (N-1) - \frac{1}{2}m(N-m)$$

for all $Gm \in [0, 1/(2N))$. In particular, $N_{\text{crit}} = 7$ is independent of the deficit angle.

Proof. The two-dimensional Green’s function is conformally invariant: for any conformal metric $g = e^{2u}|dz|^2$ on the plane, $\Delta_g G = \delta$ gives $G(z, w) = -(2\pi)^{-1} \log |z-w|$ in conformal coordinates, independent of u . Therefore the interaction energy $H = -\sum_{j < k} \log |z_j - z_k|$ is the same in conformal coordinates as on the flat plane.

The $\mathbb{Z}_N$ symmetry maps the N -gon to a regular N -gon of radius R_f in the developed plane. The angular interaction $-\sum_{j < k} \log |e^{i\theta_j} - e^{i\theta_k}|$ is independent of the radius (since $\log R$ drops from the Hessian), and so are the Fourier eigenvalues $m(N-m)/2$ of the tangential Hessian. The constraint curvature contributes $+(N-1)$ independently of R_f , completing the Havelock decomposition. $\square$

The topology change at $\beta = 1/N$. At the Gauss–Bonnet endpoint $Gm = 1/(2N)$, the total deficit exhausts the full curvature budget and the spatial slice *compactifies* to S^2 . On the sphere, the Green’s function acquires a curvature correction: $G^{S^2} = -(2\pi)^{-1} \log |z-w| + (4\pi)^{-1} \log[(1+|z|^2)(1+|w|^2)]$ (in stereographic coordinates). The additional term replaces $C_1 = N-1$ with $C_1 = (N-1) \cos \varphi < N-1$, giving $N_{\text{crit}}^{S^2} \leq 6$. The transition from $N_{\text{crit}} = 7$ (flat, all $Gm < 1/(2N)$) to $N_{\text{crit}} \leq 6$ (sphere) is therefore *discontinuous*: it occurs at the topology change, not through a gradual backreaction. The curvature-stability dichotomy of Proposition 13.3 arises from the cosmological constant Λ (which sets $\xi = R^2/\ell^2$), not from the deficit angles.

A remark on “no gravitational force in 2+1D.” Pure 2+1D gravity has no propagating degrees of freedom and no Newtonian $1/r$ force at a distance. But point masses on an AdS_3 spatial slice $\Gamma \setminus \mathbf{H}^2$ do have a well-defined static potential: the Green’s function of the Laplacian, which is logarithmic. This potential does not propagate dynamically

(there are no gravitational waves), but it determines the equilibrium energy landscape. The stability analysis depends only on the second variation of this static energy, for which the Havelock decomposition applies exactly as in the vortex case.

Proposition 13.1 transfers the full palindromic hierarchy to gravity:

- $N_{\text{crit}} = 7$ on any spatial slice (universality, Paper I, Theorem 6.8).
- The palindromic thresholds of Paper I, §6 are *gravitational phase transitions*: algebraic-number curvature values at which the stability of the mass polygon changes.
- The five algebraic-integer thresholds ($\xi^*(8) = 8-3\sqrt{7}$, ..., $\xi^*(23) = \varphi^{-2}$) give parameter-free predictions for the stability of mass rings in 2+1D AdS.

13.1 The gravitational prediction

The results of this section yield a concrete, parameter-free prediction for 2+1D gravity.

Proposition 13.3 (Maximum stable polygon in 2+1D gravity). *In linearised 2+1D gravity with cosmological constant Λ ($Gm \ll 1/(4N)$, weak-field regime), the maximum number of equal-mass conical singularities that can form a stable regular polygon on a spatial slice satisfies*

$$\text{sgn}(N_{\text{crit}} - 7) = -\text{sgn}(\Lambda).$$

Specifically: $N_{\text{crit}} = 7$ for $\Lambda = 0$; $N_{\text{crit}} \leq 6$ for $\Lambda > 0$; and $N_{\text{crit}} \geq 8$ for $\Lambda < 0$, with N_{crit} non-decreasing in $|\Lambda|$ (Proposition 5.1). The transitions occur at the palindromic thresholds $\xi^(N)$ of Paper I, §5, with $\xi = R^2/\ell_{\text{AdS}}^2$.*

Proof. The gravitational Havelock decomposition (Proposition 13.1) gives $\lambda_m = C_1(\xi) - m(N-m)/2$. The stability condition $\lambda_m > 0$ for all m , the monotonicity of $C_1(\xi)$, and the curvature-stability dichotomy (Paper I, Proposition 6.9) transfer directly. $\square$

Validity. The weak-field condition and the stability condition are *independent*: Gm controls the deficit angle (and hence the validity of the logarithmic approximation), while $\xi = R^2/\ell^2$ controls the ring position (and hence the stability). At 1% of the maximum mass ($Gm = 0.01/(4N)$): the total deficit angle is $\approx 3.6^\circ$ for all N , well within the logarithmic regime. The constraint structures (angular momentum in gravity, angular impulse in vortex dynamics) are isomorphic to $O(Gm)$: both arise from the $SO(2)$ rotational symmetry of the polygon, and the relativistic corrections to the gravitational angular momentum shift the Havelock eigenvalue by $O(Gm) \sim 1\%$ —negligible against binding eigenvalues of $O(1)$.

Code availability. All numerical verifications are in the companion repository (<https://github.com/gspea/planetary-polygons-unified>). Parameter accounting and predictions appear in Paper VI.

A Infinite-dimensional convergence proof

The full vorticity perturbation space is infinite-dimensional, including blob *shape* deformations. The Hessian of the 2D Euler energy $E[\omega] = \frac{1}{2} \iint \omega(x)G(x,y)\omega(y) dx dy$ at $\omega_\varepsilon = \sum_k \kappa \phi_\varepsilon(\cdot - z_k)$ decomposes into three sectors:

(a) *Center translations.* The Hessian restricted to the $2N$ -dimensional subspace of center displacements is *exactly* the blob Hessian of Proposition 7.1. Convergence to the point vortex Hessian at $O(\varepsilon^\alpha)$ follows from the convolution estimates.

(b) *Shape perturbations.* Mean-zero, center-preserving deformations enter through the self-energy. The second variation at mean-zero $\delta\omega$ is:

$$\delta^2 E_{\text{self}} = \frac{1}{4\pi^2} \|\delta\omega\|_{H^{-1}}^2. \quad (\text{A.1})$$

The rescaling $\phi_\varepsilon(x) = \varepsilon^{-2}\phi(x/\varepsilon)$ gives $E_{\text{self}}[\phi_\varepsilon] = c/\varepsilon^2$, so shape modes are *unconditionally stable*.

(c) *Center-shape coupling.* The vanishing monopole and dipole moments of $\delta\psi$ give quadrupole coupling: $C_{jk} = O(\varepsilon^2/d_{\min}^3)$ (numerically verified for the Cauchy blob at $d_{\min} \geq 3\varepsilon$). The ratio to the shape eigenvalue is $O(\varepsilon^4/d_{\min}^3) \rightarrow 0$ as $\varepsilon \rightarrow 0$.

Proposition A.1 (Inertia preservation). *For $\varepsilon < \varepsilon_0(N)$, where $\varepsilon_0 = (\lambda_{\min} d_{\min}^6)^{1/4}$, the full Hessian of $E[\omega_\varepsilon] + \mu_L L[\omega_\varepsilon]$ on the space of $\mathbb{Z}_N$ -symmetric vorticity perturbations has the same inertia as the point vortex Lagrangian Hessian of Theorem 3.2.*

Proof. Write the constrained full Hessian in block form $\mathcal{H} = \begin{pmatrix} H_{cc} & H_{cs} \\ H_{cs}^\top & H_{ss} \end{pmatrix}$. By (A.1), $\delta^2 E_{\text{self}} = (1/4\pi^2)\|\delta\omega\|_{H^{-1}}^2$. On the disk of radius ε , the Poincaré inequality $\|\delta\omega\|_{H^{-1}} \geq (c_P/\varepsilon)\|\delta\omega\|_{L^2}$ for mean-zero perturbations gives $H_{ss} \geq (c_P^2/(4\pi^2\varepsilon^2))\text{Id}$, hence H_{ss} is coercive. By Proposition 7.1, $\lambda_c^{(k)}(\varepsilon) = \lambda_k^{\text{Hav}} + O(\varepsilon^2)$. From (c), $\|H_{cs}\|_{\text{op}} = O(\varepsilon^2/d_{\min}^3)$. By the Schur complement, the correction is $O(\varepsilon^6/d_{\min}^6) \rightarrow 0$, preserving signs. $\square$

The explicit concentration condition is

$$\frac{\varepsilon}{d_{\min}} < \delta(N) := \sqrt{\frac{\lambda_{\min}(N)}{6(2N-3)}}, \quad (\text{A.2})$$

giving $\delta(4) = 0.18$, $\delta(5) = 0.15$, $\delta(6) = 0.096$.

Corollary A.2 (Inertia preservation for $N \leq 6$). *For $N \leq 6$ and $\varepsilon/d_{\min} < \delta(N)$, the full Hessian inertia equals the Havelock inertia.*

Corollary A.3 (Marginal-mode resolution for $N = 7$). *For $N = 7$, $\delta(7) = 0$, but $\lambda_{\min}(\varepsilon) > 0$ for all $\varepsilon \in (0, d_{\min})$: in Region 1 ($\varepsilon < 2/\sqrt{|\mathcal{C}'_7|}$), $\lambda \geq \varepsilon^2(4 - |\mathcal{C}'_7|\varepsilon^2) > 0$; in Region 2, interval arithmetic on a 2,000-point grid gives $\lambda \geq 0.34 > 0$.*

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